Chapter 1: cumulativity from the perspective of homogeneity

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1 What is cumulativity?

When two or more plural-referring expressions[1] are arguments of the same verb, they give rise to the so-called cumulative reading. In (1), the cumulative reading asserts that the squirrels and the nuts were involved in some cracking but does not specifically say which of the squirrels cracked which of the nuts. A good enough paraphrase could be (1b)[2]:

(1) a. The 10 squirrels cracked the 15 nuts
    b. **Truth-conditions:**
    Every squirrel cracked a nut.
    Every nut was cracked by a squirrel.

The presence of these readings is not, by itself, a semantic puzzle. It could be analyzed as a simple predication, as in (2). Any inference drawn from (1a), under that view, could be attributed to the meaning of the word *crack* and what it means for a plurality of squirrels to crack a plurality of nuts. This type of analysis is very plausible and has been pursued in Roberts (1987); Scha (1984).

(2) a. $[\![\text{cracked}]\!]\,(\iota\text{10-squirrels})(\iota\text{15-nuts})$
    b. **Lexical stipulation:**
    $[\![\text{cracked}]\!]\,(X)(Y)$ iff
    every one of X cracked one of Y
    and every one of Y cracked one of X

However, more complicated examples by Schein (1993) show that at least sometimes, lexical stipulations alone cannot deliver attested cumulative readings. Take the sentence in (3a) for instance ; it simply replaces one of the plural-referring expressions with a quantifier *every*. It

[1] In this thesis, the technical words or mathematical operators are underlined on first mention and link to a glossary at the end.

[2] This paraphrase, found in Scha (1984) but made prominent in Sternefeld (1998) isn't adequate when collective actions are possible, for instance if more than one squirrel collaborate to crack one nut. This chapter ignores collectivity for the time being, which we turn back to in chapter 2.

seems that this sentence can be paraphrased similarly to the original sentence in (1) (cf (3b)[3]): it is true if every squirrel cracked a nut and every nut was cracked by a squirrel. Given the universal quantificational semantics commonly attributed to *every*, we may expect that the meaning of (3a) would be rendered as in (3d). Along with the lexical stipulation of (2b), that would mean that the sentence could be paraphrased as in the second line of (3d):

(3) a. The ten squirrels cracked every nut.
b. **Truth-conditions:**
Every squirrel cracked a nut.
Every nut was cracked by a squirrel.
c. $\forall x \in [\![\text{nut}]\!], [\![\text{cracked}]\!](\iota 10\text{-squirrels})(x)$
d. **Predicted:**
for every nut x*, the squirrels cracked* x
$\leftrightarrow$*for every nut* x*, every squirrel cracked* x *and* x *was cracked by every squirrel*

Problematically, this is not the cumulative reading that the sentence seems to have. At least one of our assumptions is to blame: either the cumulative reading of that sentence does not arise through lexical stipulations or our assumptions about *every* need to be revised or both.

One way to revise our stipulation for *every* would be to claim that in some circumstances at least, *every nut* denotes the same object that *the fifteen nuts* denotes, a plurality of fifteen nuts. Since, as we saw in (1), there is no particular problem in accounting for ordinary cumulative sentence with meaning postulates if both the subject and the object denote pluralities, (3) would not pose any particular issues under this assumption. However, with example (4), Schein (1993) has established that even in its cumulative reading, *every* retains a distributive semantics for elements in its scope, i.e. there are four new plays *per* player. If *every nut* denoted a plurality of quarterbacks, this obligatory distributive semantics would be unexpected ; (5), by contrast, does not require *the quarterbacks* to be read distributively.

(4) a. The ten video-games taught every quarterback four new plays.
b. **Truth-conditions:**
Every quarterback was taught two new plays by some of the video-games
Every video-game taught a quarterback some plays.

(5) The ten video-games taught the quarterbacks four new plays.
(ok) four new plays in total

All of this suggests that putative group readings of *every* are not the reason behind cumulative readings of *every*. Since *every* cannot be blamed for generating cumulative readings, we can conclude that cumulative readings sometimes arise compositionally, rather than through lexical stipulations. This conclusion, drawn in various works, has been most prominently defended by Beck and Sauerland (2000)[4]. The goal of this chapter is to build a theory of the

[3]There are some truth-conditional differences between this type of sentence and the ordinary cumulative sentences like (1): one involves collective action and is covered in chapter 2, the other negative sentences and is discussed in section VII.4.

[4]On the basis of different data ; we review their approach in chap. 3.

composition which can account for cumulative readings and, in particular, cumulative readings of *every*.

Other quantifiers than *every* Cumulativity is a very general phenomenon and *every* is not an exception among quantifiers in giving rise to cumulative readings. The examples in (6-8) are all examples of cumulative readings with varied quantifiers.

(6) a. The ten squirrels cracked between 28 and 35 nuts.
b. **Truth-conditions:**
Between 28 and 35 nuts were cracked by some of the ten squirrels.
Every one of the ten squirrels cracked a nut.

(7) a. The ten squirrels cracked a prime number of nuts.
b. **Truth-conditions:**
A prime number of nuts were cracked by some of the ten squirrels.
Every one of the ten squirrels cracked a nut.

(8) a. The ten squirrels cracked most nuts.
b. **Truth-conditions:**
A prime number of nuts were cracked by some of the ten squirrels.
Every one of the ten squirrels cracked a nut.

However, the examples above are less interesting to the study of cumulative readings ; unlike *every*, accounting for their truth-conditions doesn't seem to require resources beyond the meaning postulate already mentioned in (2b).

While their intrinsic interest to the theorist may differ, this is not ground to consider cumulative readings of *every* and cumulative readings of other quantifiers separate cases. As we will see, both types of readings satisfy identical generalizations: both obey the generalizations discussed in section II.2 and both exhibit the subject/object asymmetries discussed in section VII.3. This is why I will also include the study of other quantifiers in the scope of my analysis.

This chapter's dataset. There is a general phenomenon of cumulative readings, as this section hopes to have shown. Some of the varied cases of cumulativity can be reduced to a simple lexical stipulation on the meanings of verbs. If it were for them alone, our semantic work would be done. However, cumulative readings of *every* and in particular, the video-game examples show that at least some of these cases must arise through compositional means.

In addition to setting the problem, this section serves the purpose of showcasing the core dataset of this chapter. In the end, the analysis will aim to account for all of the data points in (9):

(9) a. The 10 squirrels cracked the 15 nuts
b. The ten squirrels cracked every nut.
c. The ten video-games taught every quarterback four new plays.
d. The ten squirrels cracked between 28 and 35 nuts.
e. The ten squirrels cracked a prime number of nuts.

For most of this chapter however, ordinary cumulative sentence, like (9a), will be left aside. We will focus on establishing generalizations and the theory on examples in b-d. This is because all of the generalizations we establish on b-d, while also valid for ordinary cumulative sentence, are less easily seen with them. Only in section VII.4 will we be able to give a fair treatment of (9a).

Throughout this chapter, I will disregard the possibility of collective action. In particular, I will assume that the squirrels of our examples don't team up to crack nuts. Similarly, video-games always teach full plays ; it doesn't take a quarterback more than one video-game to learn a play. Insofar as these assumptions hold, our account will make the correct predictions. The necessary amendments for the collective case when these assumptions do not hold will be given in chapter 2.

Roadmap. This chapter has three parts. The first part, which consists of section II, studies cumulativity and another phenomenon related to plurality, homogeneity. From this dual perspective, section II arrives at two descriptive generalizations, the *low-scope existential* generalization and the *exhaustive participation* generalization. Together, these generalizations completely characterize the phenomenon of homogeneity and cumulativity in our dataset.

In the second part, which consists of section III, I give a formal account of the first generalization, the low-scope existential generalization. I propose that verbs come to the syntax with built-in existential meanings. This proposed mechanism of "*built-in existentials*" will be shown to be independently attested in disparate semantic phenomena, such as *impersonal pronouns* or *metonymy*.

The third part finalizes the theory by providing an account of the *exhaustive participation* inferences. In section IV, I show that *exhaustive participation inferences* of non-cumulative sentences are similar to Free Choice inferences (following Bar-Lev (2018b)) and *exhaustive participation inferences* of cumulative sentences are similar to distributive implicatures. Sections V and VI open a parenthesis on Free Choice and distributive implicatures: section V describes two accounts of Free Choice implicatures, innocent inclusion and recursive exhaustification ; section VI shows that only one of these accounts - the recursive exhaustification of Fox (2007) - can truly capture distributive implicatures. Closing the parenthesis on scalar implicatures, section VII applies the theory to our dataset: not only is our dataset fully captured by the theory, including the more complex video-game example, it also provides a principled account of asymmetries in cumulative reading of *every* (Champollion, 2010; Haslinger and Schmitt, 2018; Kratzer, 2000).

II Generalizations about plural interpretations: the dual perspective of homogeneity and cumulativity

The previous section has established cumulativity as a semantic puzzle. However, my description of the facts is still partial. In this section, we will see that there is much to be learned about cumulative readings from the perspective of homogeneity, another aspect of plural interpretation. Specifically, we will systematically compare sentenes with cumulative readings with their negation. By paying close attention to patterns that arise from this comparison, this section will establish two descriptive generalizations about the interpretation of plural-referring expressions, which together completely characterize the truth-conditions of cumulative sentences. It will be the goal of the rest of the chapter to provide an account of these inferences.

II.1 Homogeneity and why it relates to cumulativity

There are situations which neither a sentence containing a plural nor its negation can appropriately describe. For instance, in the context of (10), neither (10a) nor (10b) can be truthfully asserted.

(10) **Context:** *Half of the dancers are smiling and the other half is crying*

a. # The ten dancers are smiling.

b. # The ten dancers aren't smiling.

This "*truth-value gap*" that plural-referring expressions give rise to is what we call *homogeneity*[5] (Bar-Lev, 2018a,b; Kriz, 2015; Križ, 2016; Križ and Spector, 2017; Löbner, 2000; Magri, 2014; Malamud, 2012; Schwarzschild, 1993). To be more precise, *the ten dancers* in the positive sentence (11a) seems to have a universal or quasi-universal meaning, being roughly equivalent to *every dancer*, whereas it only has an existential meaning in the negative sentence *one of the dancers*.

(11) a. The ten dancers smiled.
⇜⇝ ***every*** *one of the ten dancers smiled*

b. The ten dancers didn't smile.
⇜⇝ *it's not the case that* ***one*** *of the ten dancers smiled*

The relevance of this phenomenon to the puzzle of cumulativity may not seem obvious. However, we are now going to see that there are parallels between the way homogeneity and cumulativity have been described in the literature. These parallels will be our initial motivation to study both phenomena in parallel.

Existentials for homogeneity and cumulativity. According to one type of analysis (Bar-Lev, 2018b; Magri, 2014) of the truth-value gap exhibited in (11a) and (11b), the meaning of plural-referring expression, as revealed in negative environments, is underlyingly existential. This means that by default, plural-referring expression receive the same existential readings

[5] By extension, the name is also applied to truth-value gaps exhibited by other constructions such as conditionals (Bassi and Bar-Lev, 2018), embedded questions(Kriz et al., 2015).

seen in (11b). In positive environments, they propose that the underlying existential reading is masked by additional inferences, which come about through exhaustification (additional inferences color-coded in gray below).

(12) a. The ten dancers smiled.
$\rightsquigarrow \exists x \underset{\cdot\cdot}{<} \iota 10\text{-dancers}, \mathsf{smiled}(x) \wedge \forall x < \iota 10\text{-dancers}, \mathsf{smiled}(x)$

b. The ten dancers smiled.
$\rightsquigarrow$
$\neg \exists x < \iota 10\text{-dancers}, \mathsf{smiled}(x)$

The interesting connection enters the stage at this point: independently from any debates around homogeneity, Bayer (2013) has also suggested that plural-referring expressions may sometimes receive existential interpretations, precisely in order to account for cumulative readings of *every*. Specifically, he observes that treating *the squirrels* as an existential (cf red quantifier in (13b)), we derive one of the two inferences that form the truth-conditions of cumulative reading, namely that *every nut was cracked by a squirrel*. He notes that the other inference *every squirrel cracked a nut*, color-coded in gray below, is missing. He refers to this problem as *the leakage problem*.

(13) a. The ten squirrels cracked every nut.
b. **Bayer's translation:** $\forall x \in \mathsf{nut}, \exists y < \iota 10\text{-squirrels}, \mathsf{crack}(x)(y)$
c. **Attested truth-conditions:**
Every nut was cracked by a squirrel.
Every one of the ten squirrels cracked a nut.

This "*leakage*" problem is parallel to the situation discussed in (11) for homogeneity: with homogeneity, additional inferences were missing in order to account for the attested universal meaning of plural-referring expression in positive environment. As we saw, these additional inferences were only required for positive sentences. The new observation I make here is that the same holds of the cumulative sentence in (13): the negation of Bayer's paraphrase is perfectly adequate for negative sentences ; no additional inferences is needed to derive the meaning of the sentence in negative environment. Indeed, consider the negative version of (13). Speakers judge that this sentence is true if and only not every nut was cracked by a squirrel, i.e. the negation of Bayer's paraphrase in (13b). Whether or not all squirrels cracked a nut is deemed irrelevant to the truth of the sentence. No "*leakage*" problem arises in negative environments.

(14) a. The ten squirrels didn't crack every nut.
b. **Bayer's translation:** $\neg \forall x \in \mathsf{nut}, \exists y < \iota 10\text{-squirrels}, \mathsf{crack}(x)(y)$
c. **Attested truth-conditions:**
Not every nut was cracked by a squirrel.

Summary. Conjoining two independent lines of thought in the literature, we see two parallels between cumulativity and homogeneity: first, both give rise to paraphrases where the plural-referring expressions are glossed as existentials ; second, these paraphrases must be supplemented with additional inferences for to fully capture the truth-conditions of positive

sentences. The next section is devoted to spelling out more precisely what the parallels consist in. I present two descriptive generalizations which together completely characterize the phenomena of homogeneity and cumulativity in our dataset.

II.2 The two generalizations

Let us reflect on the two examples seen so far. They are repeated below in their positive and negative form.

(15) a. The dancers smiled
b. $\underline{\exists x \prec \iota 10\text{-dancers},}\ \mathsf{smiled}(x)$
$\forall x \prec \iota 10\text{-dancers},\ \mathsf{smiled}(x)$

(16) a. The dancers didn't smile
b. $\neg\underline{\exists x \prec \iota 10\text{-dancers},}\ \mathsf{smiled}(x)$

(17) a. The squirrels cracked every nut.
b. $\forall x \in \mathsf{nut}, \underline{\exists y \prec \iota 10\text{-squirrels}}, \mathrm{crack}(x)(y)$
$\forall y \prec \iota 10\text{-squirrels}, \exists y \in [\![\mathrm{nut}]\!], \mathrm{crack}(x)(y)$

(18) a. The squirrels didn't crack every nut.
b. $\neg\forall x \in \mathsf{nut}, \underline{\exists y \prec \iota 10\text{-squirrels}}, \mathrm{crack}(x)(y)$

As seen in the previous section, in all of the paraphrases of the negative sentences and at least one half of the paraphrase of the paraphrase of positive sentences, the plural-referring expression is translated as a existential over parts of the plurality denoted by plural-referring expression (underlined in the examples above). We may moreover observe that when there are operators in the clause, this existential always takes the lowest scope within the paraphrase. Any other scope-ordering in our logical paraphrases misses the attested reading:

(19) a. The dancers didn't smile
b. **Inadequate paraphrase**
$\underline{\exists x \prec \iota 10\text{-dancers},}\ \neg\mathsf{smiled}(x)$

(20) a. The squirrels didn't crack every nut.
b. **Inadequate paraphrase**
$\neg\underline{\exists y \prec \iota 10\text{-squirrels}}, \forall x \in \mathsf{nut}, \mathrm{crack}(x)(y)$
= *no squirrel cracked every nut*

This makes the basis for the first generalization[6]:

[6] The generalization specifies that the existential corresponding to the plural-referring expression must scope lower than any element *in the clause*. The addition of "*in the clause*" does not rule out the existential scoping higher than an operator in an embedded clause, as in (21):

Low-scope existential meaning

In the logical paraphrase of a sentence, a plural-referring expression are interpreted as an existential over parts which takes a lower scope than any other quantifier or operator in the clause:

$$\text{the squirrels} \rightsquigarrow \ldots Qy, \ldots \exists x \prec \iota\mathsf{squirrels}, \ldots$$

The second generalization we can make from our sample so far concerns the shape of the inferences which are only attested in positive sentences (in gray above). They all assert that all members denoted by the plural-referring expression participated in the action described by the verb. What it means for an x to "*participate in the action described by the verb*" in our paraphrases is simply to be true of the predicate with all of the other arguments existentially bound by appropriately restricted quantifiers. More formally, we can state the generalization as follows[7]:

Exhaustive participation inferences

In positive environments only, each plural-referring expression gives rise to an *exhaustive participation inference*.
This *exhaustive participation inference* of a plural-referring expressions DP is of the form "$\forall x \prec [\![DP]\!], \exists y \in C, \ldots$" where the plural-referring expression is interpreted universally with high-scope and the other arguments are interpreted as existentials ranging over the NP restriction of the arguments, as below:

$$\text{the NP} \ldots \mathcal{Q}\ \text{NP}' \ldots \mathcal{Q}\ \text{NP}'' \ldots \rightsquigarrow \forall x \prec [\![\text{NP}]\!], \exists y \in [\![\text{NP}']\!], \exists z \in [\![\text{NP}'']\!], \ldots$$

There are several comments to make about this generalization. First, the fact that this condition is specified to apply *only* in positive environments means that plural-referring expressions will yield truth-value gaps, just as we observed. Indeed, there will be inferences of positive sentences which are not found under negation in negative sentences. The second remark is a forewarning: because of cases discussed in section VII.2, we will see that this generalization is not true in all generality, in ways that the analysis predicts. However, it is enough as far as our main dataset is concerned.

In the remainder of this section, we will put these generalizations to the test by seeing how they apply beyond the two classes of examples that motivated them. We will turn to the other sentences in the dataset: Schein's video-game example, cumulative readings with other

(21) The headlines didn't say the president wasn't in trouble.
$\neg\exists x \prec \iota\mathsf{headlines}, \mathsf{say}(x)(\neg\mathsf{in\text{-}trouble}(\iota\mathsf{president}))$

On closer inspection however, note that the addition of *in the clause* is in fact necessary since a quantifier may not scope lower than the variable it binds. Since its variable is in the matrix clause, the existential cannot scope below than point.

[7] The informed reader will not have failed to recognize that the shape of the exhaustive participation inferences is very much comparable to Schein (1993)'s own analysis of cumulative readings. Exploring the parallels between the generalizations established in previous literature is the object of chapter 3.

quantifiers than *every* and finally a non-cumulative sentence.

The video-game examples In Schein (1993)'s video-game example, a plural-referring expression shares the predicate with *every* and a numeral quantifier. The low-scope existential generalization states that one half of the truth-conditions of the video-game example can be obtained by translating the plural-referring expression as a low-scoping existential, as in the black part of (22b). The other half consists in the *exhaustive participation inference* of *the ten video-games*. From the second generalization, we know that this exhaustive participation inference will be as the gray part of (22b). We can now compare the paraphrase we obtained by following the mandates of our two generalizations to the actual reading of the sentence already given in section I and repeated below: the match is perfect.

(22) a. The ten video-games taught every quarterback two new plays.

b. $\forall y \in \mathsf{quarterback}, \exists 2 z \in \mathsf{plays}, \exists x \prec \iota\mathsf{video\text{-}games},\ \mathsf{teach}(x)(y)(z)$
$\forall x \prec \iota\mathsf{video\text{-}games}, \exists y \in \mathsf{quarterback}, \exists z \in \mathsf{plays},\ \mathsf{teach}(x)(y)(z)$

c. **Attested reading:**
Every quarterback was taught two new plays by some of the video-games
Every video-game taught a quarterback some plays.

Our generalizations also leads us to the expectation that sentence (22a) has a truth-value gap. Namely, its negation should only be the negation of the black part of (22b) rather than the negation of the whole of (22b), since the *exhaustive participation inference* is only derived for positive contexts. In other words, (23) should mean (24a) rather than (24b).

(23) The ten video-games didn't teach every quarterback two new plays.

(24) a. **Expected truth-conditions:**
$\neg\forall y \in \mathsf{quarterback}, \exists 2 z \in \mathsf{plays}, \exists x \prec \iota\mathsf{video\text{-}games},\ \mathsf{teach}(x)(y)(z)$

b. **Unexpected truth-conditions:**
$\neg\forall y \in \mathsf{quarterback}, \exists 2 z \in \mathsf{plays}, \exists x \prec \iota\mathsf{video\text{-}games},\ \mathsf{teach}(x)(y)(z)$
$\vee\neg\forall x \prec \iota\mathsf{video\text{-}games}, \exists y \in \mathsf{quarterback}, \exists z \in \mathsf{plays},\ \mathsf{teach}(x)(y)(z)$

One way to probe this difference on native speakers is to compare the acceptability of continuations provided as justifications for the statement:

(25) The ten video-games didn't teach every quarterback two new plays.

a. #…indeed, although all quarterback learned two plays, one of the video-games wasn't working.

b. …indeed, although all video-games taught some plays, one quarterback didn't learn anything.

All in all, the generalizations extend to the relatively complex case of video-game examples, which is a testament to their robustness.

Cumulative reading with other quantifiers than *every*. As we noted, cumulative sentences can be constructed with other quantifiers than *every*.

(26) a. The ten squirrels cracked at least 5 nuts.
b. The ten squirrels cracked most of the nuts.

As before, our paraphrases have two parts: a core meaning and an *exhaustive participation inference*. By the low-scope existential paraphrase, *the ten squirrels* must be translated in the paraphrase as a low-scope existential meaning ; this is what I have done in the black part on (27). To obtain the exhaustive participation inference, as per the second generalization, one simply needs replaces every argument other than *the ten squirrels* by an existential and treat the latter as a high-scope universal. This is done in the gray part of (27). The truth-conditions predicted by the two generalizations match the observed truth-conditions.

(27) a. **Predicted truth-conditions of (26a):**
$\#\{x \in \mathsf{nut} \mid \exists y \prec \iota\mathsf{squirrels}, \mathsf{cracked}(x)(y)\} > 5$
$\wedge \forall x \prec \iota\mathsf{squirrels}, \exists y \in \mathsf{nut}, \mathsf{crack}(x)(y)$
b. **Predicted truth-conditions of (26b):**
$\frac{\#\{x \in \mathsf{nut} \mid \exists y \prec \iota\mathsf{squirrels}, \mathsf{cracked}(x)(y)\}}{\#\mathsf{nut}} > .5$
$\wedge \forall x \prec \iota\mathsf{squirrels}, \exists y \in \mathsf{nut}, \mathsf{crack}(x)(y)$

What we see here is that from the perspective of the generalizations, nothing really distinguishes these cases from the video-game example and the cumulative sentence with *every*. My approach thus departs from some of the previous literature (reviewed in chapter 3) which has frequently given the latter two examples a distinguished status.

The low-scope existential generalization in non-cumulative sentences So far, all of the examples used to showcase the generalizations have involved cumulative sentences. However, the generalizations have been stated to hold for plural interpretation in general. In principle then, their empirical use could also be appreciated on non-cumulative sentences as well.

Consider (28). It only contains one plural-referring expression and *only*. As such, it does fall within what I narrowly defined as a cumulative reading. Let us focus on the assertive meaning of the uttered

(28) Last week, the neighbor's dogs only barked on Sunday

The low-scope generalization predicts that one part of (28) is paraphrasable as the black part of (29). This part asserts that on days that were not Sunday, no dog barked. Whether the *exhaustive participation* generalization applies is more tricky to say, since *only* does not form a typical positive or negative environment. However, whether it applies or not, the derived inference in gray below does not strengthen the assertive meaning of the sentence. The predicted reading match the intuition that the sentence can only be true if no barking happen on any other day than Sunday.

(29) $\neg\exists d \neq \mathsf{Sunday}, \exists y \prec \iota\mathsf{10\text{-}dogs}, \mathsf{bark\text{-}on}(y)(d)$
$\wedge \forall y \prec \iota\mathsf{10\text{-}dogs}, \neg\exists d \neq \mathsf{Sunday}, \mathsf{bark\text{-}on}(y)(d)$

Note what would happen for (28), if the *low-scope existential* generalization did not specify that the existential had to be low-scope. If the *exhaustive participation inference* generaliza-

tion does apply, nothing would change since the exhaustive participation inference already delivers the right meaning:

(30) $\exists y \prec \iota\text{10-dogs}, \neg\exists d \neq \text{Sunday}, \text{bark-on}(y)(d)$
$\wedge \forall y \prec \iota\text{10-dogs}, \neg\exists d \neq \text{Sunday}, \text{bark-on}(y)(d)$
= *no dog barked on any other day than Sunday*

However, there is reason to doubt that the meaning of (28) is due to an exhaustive participation inference. This is because we find no truth-value gap as far as the assertive meaning is concerned. If (30) truly represents the truth-conditions of (28), then we would expect (31a), its negation, to receive the meaning in (31b). In fact, it receives the meaning of (31c), which is simply the negation of the truth-conditions specified in (29).

(31) a. I doubt that the neighbor's dogs only barked on Sunday.
b. *high-scope existential:*
$\neg\exists y \prec \iota\text{10-dogs}, \neg\exists d \neq \text{Sunday}, \text{bark-on}(y)(d)$
$\leftrightarrow \forall y \prec \iota\text{10-dogs}, \exists d \neq \text{Sunday}, \text{bark-on}(y)(d)$
=*all dogs barked on some other day*
c. *low-scope existential:*
$\neg\neg\exists d \neq \text{Sunday}, \exists y \prec \iota\text{10-dogs}, \text{bark-on}(y)(d)$
$\leftrightarrow \exists y \prec \iota\text{10-dogs}, \exists d \neq \text{Sunday}, \text{bark-on}(y)(d)$
=*some dogs barked on some other day*

This example both confirms the validity of the generalizations on non-cumulative sentences, in addition to illustrating once more the importance of the *low-scope* property. We will come back to this example in section III.

II.3 Outlook

Both the analysis of cumulativity and homogeneity suggest that plural receive an existential interpretation, as independently suggested by Bar-Lev (2018b); Magri (2014) for homogeneity and Bayer (2013) for cumulativity. Here, I furthermore suggested that this existential interpretation is bound to take the lowest-scope in the clause. These facts formed the basis of our first generalization, the low-scope existential generalization.

In positive environments, this existential reading may be obscured by additional inferences, the *exhaustive participation inferences*, which asserts that all individuals denoted by plural-referring expression are in some sense involved in the action. It is the object of the second generalization to characterize the shape of the *exhaustive participation inferences*.

In the next section, we will go beyond the descriptive level and I will show how both generalizations can be rendered formally. We will analyze the two generalizations of the previous section in two separate sections. In section III, I will propose that verbal predicates build in existential meanings for their plural arguments ; this will provide an account of the *low-scope existential generalization*. While strange, this proposal will have unexpected parallels in other areas of the semantics. Section IV starts our investigation of *exhaustive participation inferences*. Expanding on Bar-Lev (2018b), who proposed that *exhaustive participation inferences*

(in my terminology) are free-choice-like inferences, I will propose that *exhaustive participation inferences* are a case of *generalized distributive implicatures*, of which free-choice inferences are a sub-case. I will then show the mechanism of recursive exhaustification by which both can be derived.

III Analysis of the low-scope existential meaning

III.1 First try: a problematic account

An operator existential. To ensure that plural-referring expressions are interpreted as existential quantifiers, we could simply assume that every plural-referring expression must be accompanied at LF with an operator $\exists$ which effectively converts it into an existential quantifier (over singularities, since we are still disregarding collective action, which will be addressed in chapter 2).

(32) a. $[\![\exists]\!] = \lambda X_e.\lambda p_{et}.\, \exists x \prec X, p(x)$
b. [The dancers $\exists$] smiled
$\rightsquigarrow \exists x \prec \iota\mathsf{dancers}, \mathsf{smiled}(x)$

Problematically, this analysis predicts that the scope of the existential quantification is married to the position of the plural-referring expression[8]. If, as per our generalization, this existential quantification always takes the lowest scope within the clause, then by this analysis, it must be that the plural-referring expression is always syntactically dominated by any other operator in the clause.

Setting aside the fact that it is unclear how we might impose this syntactic restriction, the data that we have seen shows that in fact, the scope of the existential quantification seem to completely disregard scopal restrictions that usually affect the plural-referring expression's syntactic position. Consider (28), repeated below in (33).

(33) Last week, the neighbor's ten dogs only barked on Sunday.

In this sentence, the existential paraphrase of *the neighbor's ten dogs* under-scoped VP *only*. By contrast, quantifiers in that syntactic position have an exceedingly hard time under-scoping VP *only*.

(34) a. Every dog only barked on Sunday.
$\neq$ *only on Sunday did every dog bark* (*only $\gg \forall$)
b. Some dog only barked on Sunday.
$\neq$ *only on Sunday did some dog bark* (*only $\gg \exists$)
c. Half of the ten dogs only barked on Sunday.
$\neq$ *only on Sunday did half of the dogs bark* (*only $\gg \exists$)

This may not be a hard and fast rule. Some speakers (R. Schwarzschild, p.c.) do accept a low-scope for unstressed *some*, glossed below as *sm*. In this example, it is unclear to me whether

[8] This is essentially the operator that Bar-Lev (2018b) assumes. By making this assumption, he wants to identify the $\exists$ operator to the well-motivated distributivity operator (Roberts, 1987), which also exhibits homogeneity effects. However, we will have reasons not to do so, cf section VII.4 and chapter 4.

the low-scope is truly rendered possible by *sm* or by the inaccusative verb *fall*. (35b) tries to tease apart these factors.

(35) a. Sm snow only fell on Sunday.

b. Sm snow only covers Joshua. $\stackrel{?}{=}$ Only Joshua is covered by snow.

If this is not convincing, we may turn to scope-rigid languages, e.g. Hungarian[9]. In Hungarian, scope is explicitly marked by constituent order. In particular, the position of *czak* "*only*" clearly marks its scope. Unlike English, Roger Schwarzschild's examples are not replicable in Hungarian. Like English however, the existential paraphrase of a plural-referring expression like *the ten dogs* takes the lowest scope possible. This is illustrated in (36):

(36) Plurals existential meaning takes lowest scope

a. *A tiz kutyam csak Gabitot harapta meg.*
the ten dog-my only Gabito-acc bit vm

The ten dogs only bit Gabito

b. True if none of them bit Maria
False if 5 or more of them bit Maria
False if all 10 of them bit Maria
$\rightsquigarrow$ only $\gg \exists$

c. *Nem hiszem, hogy a tiz kutyam csak Gabitot harapta meg.*
not think that the ten dog-my only Gabito-acc bit vm

I don't think that the ten dogs only bit Gabito

d. False if none of them bit Maria
True if 5 or more of them bit Maria
True if all 10 of them bit Maria
$\rightsquigarrow \neg \gg$ only $\gg \exists$

Outlook. In conclusion, I hope to have shown that the scope of the existential quantifier over pluralities does not track the syntactic position of the plural-referring expression that it translates, as the operator analysis I sketched would have it.

III.2 The correct analysis

As an alternative to this view, I submit that the existential meaning is part of the meaning of the verbal predicate, as in (37):

(37) a. $[\![\exists\text{-danced}]\!] = \lambda X.\, \exists x \prec X,\ \mathsf{danced}(x)$

b. $[\![\exists\text{-cracked}]\!] = \lambda X.\lambda Y.\, \exists x \prec X, \exists y \prec Y,\ \mathsf{cracked}(x)(y)$

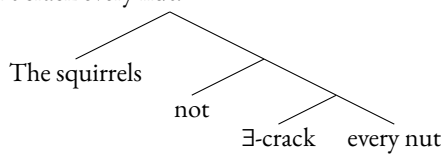
By packaging the verbal predicate with the existential semantics for plural, we semantically forbid any operator to under-scope these existentials. The scope of the existential meaning

[9] I thank Dorá Kata Takács for sharing both her language and her linguistic expertise with me.

is decoupled from the syntactic position of the argument and no syntactic stipulations constraining scope are required. We can fully appreciate these denotations on a representative sample of our dataset (introduced in section I). Since we do not yet have an account of exhaustive participation inferences, I focus on negative environment where these inferences do not arise.

(38) a. The squirrels didn't crack every nut.

b.



c. $[\![\text{every nut}]\!] = \lambda p_{eet}.\lambda X.\ \forall y \in \mathsf{nut},\ p(y)(X)$[10]

$[\![\text{cracked every nut}]\!] = \lambda X.\ \forall y \in \mathsf{nut}, \exists x \prec X, \exists y' \prec y,\ \mathsf{cracked}(x)(y')$

$\lambda X.\ \forall y \in \mathsf{nut}, \exists x \prec X,\ \mathsf{cracked}(x)(y)$

(simplification since y is atomic)

$[\![\text{not cracked every nut}]\!] = \lambda X.\ \neg \forall y \in \mathsf{nut}, \exists x \prec X,\ \mathsf{cracked}(x)(y)$

$[\![\text{the squirrels not cracked every nut}]\!] = \neg \forall y \in \mathsf{nut}, \exists x \prec \iota\mathsf{squirrels},\ \mathsf{cracked}(x)(y)$

As the reader can see, the composition is classical, the meaning of quantifiers is standard and the truth-conditions are the correct ones. Furthermore, it straightforwardly extends to the other data points of our dataset:

(39) a. The squirrels didn't crack a prime number of nuts.

b. $[\![\text{a prime number of nuts}]\!] = \lambda p_{eet}.\lambda X.\ \#\{y \in \mathsf{nut} \mid p(y)(X)\}$ is prime

$[\![\text{cracked a prime number of nuts}]\!]$

$= \lambda X.\ \#\{y \in \mathsf{nut} \mid \exists y' \prec y, \exists x \prec X,\ \mathsf{crack}(y')(x)\}$ is prime

$= \lambda X.\ \#\{y \in \mathsf{nut} \mid \exists x \prec X,\ \mathsf{crack}(y)(x)\}$ is prime

(simplification since y is atomic)

$[\![\text{not cracked a prime number of nuts}]\!]$

$= \lambda X.\ \#\{y \in \mathsf{nut} \mid \exists x \prec X,\ \mathsf{crack}(y)(x)\}$ is not prime

$[\![\text{the squirrels not cracked a prime number of nuts}]\!]$

$= \lambda X.\ \#\{y \in \mathsf{nut} \mid \exists x \prec \iota\mathsf{squirrels},\ \mathsf{crack}(y)(x)\}$ is not prime

III.3 Beyond plurality: existential shifts.

The proposal of building existential meanings into the meaning of the verb may seem strange. Interestingly, there are in fact a number of unrelated phenomena which call for a similar analysis. In many instances, natural language imposes existential meanings onto a verb's argument and this existential force takes the narrowest scope. These existential shifts include (at least): existential reading of impersonal pronouns[11], derived kind predication, and some cases of metonymy. All the phenomena are illustrated in (40).

[10] I am assuming a type-shifting approach here but of course, this is an orthogonal point: a QR-based approach would yield exactly the same result.

[11] I thank Sarah Zobel for pointing this out to me.

(40) a. **Derived Kind predication**
That kind of bee only stung Fatma.
≈ *an instance of that kind only stung Fatma*

b. **Existential readings of impersonal pronouns** (German *man*)
Gestern hat man die Uni angezündet.
yesterday has MAN the university set-on-fire

≈ "*yesterday, someone put the university on fire*" (Zobel, 2016)

c. **Physical location metonymy**[12]
NYU is in Abu Dhabi.
≈ *a physical branch of NYU is in Abu Dhabi*

Interestingly, just as I argued for plurals, the low-scope readings of these existentials always trumps independently attested scopal restrictions on their syntactic position. For instance, the existentials introduced by these shift under-scopes VP-only (or in (42), the equivalently scope-rigid *nur*)

(42) a.

b. **Derived Kind predication**
That kind of mosquito only stung Fatma.
≈ *no one beyond Fatma was stung by an instance of that kind* (only $\gg \exists$)

c. **Existential readings of impersonal pronouns** (German *man*)
Gestern hat man mich nur per E-Mail kontaktiert.
yesterday has MAN me.acc only by e-mail contacted

≈ "*yesterday, I was only contacted by e-mail*[13]"

d. **Branch metonymy**
NYU is only in Abu Dhabi.
≈ *in no other places is there an NYU branch*

The conclusion to draw from this fact is that a general mechanism of low-scope existential generation is needed not just for cumulativity, and my proposal to incorporate existentials in the meaning of the verb is the simplest way to implement this mechanism.

[12] One may wonder whether this example truly calls for an existential shift. Couldn't it just be that the predicate "*in Paris*" is true of IBM? If this were so, notes Parsons, we would predicts that (40a) and (40b) together entail (40c), by the logic of *Predicate Modification*, but they don't. Some quantificational force must prevent us from conjoining the meaning of predicates together.

(40) a. NYU is in Abu Dhabi.

b. NYU is on a hilly area.

c. $\not\Rightarrow$ NYU is in Abu Dhabi on a hilly area.

IV Exhaustive participation inferences: analogies with scalar implicature

Having established that we can ensure low-scope existential meanings for plurals by postulating that these existential meanings are packaged in the verb's denotation, our first generalization, we can turn to the second generalization, which is repeated below.

Exhaustive participation inferences

In positive environments only, each plural-referring expression gives rise to an *exhaustive participation inference*.
This *exhaustive participation inference* of a plural-referring expressions DP is of the form "$\forall x \prec [\![DP]\!], \exists y \in C, \ldots$" where the plural-referring expression is interpreted universally with high-scope and the other arguments are interpreted as existentials ranging over the NP restriction of the arguments, as below:

$$\text{the NP} \ldots \mathcal{Q}\ \text{NP}' \ldots \mathcal{Q}\ \text{NP}'' \ldots \rightsquigarrow \forall x \prec [\![\text{NP}]\!], \exists y \in [\![\text{NP}']\!], \exists z \in [\![\text{NP}'']\!], \ldots$$

This generalization will require considerably more machinery to do justice to. However, we will have guiding intuitions. In this section, I develop these guiding intuition. Specifically, we will construct an analogy between exhaustive participation inferences and certain classes of implicatures, which will guide the theory. We will see that on the one hand, the exhaustive participation inferences of *non*-cumulative sentences have similarities to Free Choice inferences (*pace* Bar-Lev (2018b)) and that on the other hand, the exhaustive participation inferences of cumulative sentences mirror distributive implicatures.

IV.1 Why implicatures?

In non-cumulative sentences, the underlying existential meaning of *the dancers* (cf (43)) is obscured by the exhaustive participation inference that all dancers smiled.

(43) a. The dancers ∃-smiled.
b. $\exists x \prec \iota\mathsf{dancers},\ \mathsf{smiled}(x)$
$\forall x \prec \iota\mathsf{dancers},\ \mathsf{smiled}(x)$

In cumulative sentences like (44), the exhaustive participation inference simply asserts that all squirrels cracked a nut:

(44) a. The squirrels ∃-cracked every nut.
b. $\forall y \in \mathsf{nut}, \exists x \prec \iota\mathsf{squirrels},\ \mathsf{cracked}(y)(x)$
$\forall x \prec \iota\mathsf{squirrels}, \exists y \in \mathsf{nut},\ \mathsf{cracked}(y)(x)$

Where could these universal inference come from? Several clues constrain our theoretical options. First, these inferences only arise in positive environments. Second, these inferences are conditioned by contextual information. This is something that I haven't commented on so far. In particular contexts, the universal force of the exhaustive participation inference

becomes a quasi-universal inference (cf (45a)) ; in some contexts, it may even be absent altogether, leaving the underlying existential meaning apparent (cf (45b)). As argued by Križ (2016); Malamud (2012), the presence of weaker meanings depends on what is relevant to the speaker *vis-á-vis* their current purposes.

(45) a. The townspeople are asleep. (Lasersohn, 1999)
$\rightsquigarrow$ *ok, if one or two professors aren't smiling*
b. The windows are open. (Malamud, 2012)
$\rightsquigarrow$ *ok, if at least one window is open*

These less-than-universal reading extend to cumulative sentences:

(46) The chickens laid every egg. (modeled after Buccola and Spector (2016))
$\rightsquigarrow$ *ok, if one or two chickens didn't lay any egg*

So exhaustive participation inferences are in general both polarity-sensitive and context-sensitive. As implicatures are also both polarity-sensitive and contingent on context, taking exhaustive participation inferences to be implicatures is a natural identification. This is the path that Bar-Lev (2018b); Magri (2014) take and that I will follow them on.

Exhaustive participation and Free Choice. Bar-Lev (2018b) specifically makes the parallel between exhaustive participation inferences and free choice inferences. Free Choice inferences (illustrated in (47) with our color-coding conventions for truth-value gaps) refers to cases where a disjunction embedded under an existential modal is interpreted as a wide-scope conjunction.

(47) a. You are allowed to eat apple or cake.
b. **Truth-conditions:**
$\Diamond(\mathsf{cake} \vee \mathsf{apple})$
$\wedge \Diamond\mathsf{cake} \wedge \Diamond\mathsf{apple}$

Bar-Lev (2018b) notes that in non-cumulative sentences too, an existential/disjunctive meaning is turned into a universal/conjunctive meaning:

(48) a. The dancers ∃-smiled.
b. $\exists x < \iota\mathsf{dancers},\ \mathsf{smiled}(x)$
$\forall x < \iota\mathsf{dancers},\ \mathsf{smiled}(x)$

Both classes of inferences - Free Choice and exhaustive participation - apply to an existential or disjunctive meaning and convert it to a universal or conjunctive meaning. This analogy is somewhat loose, since the typical free choice inference only occurs in the scope of a possibility modal (cf (47)), and the exhaustive participation in non-cumulative sentences typically occurs in the absence of a modal.

However, Bar-Lev (2018b) argues, following a substantive previous literature, that the free choice exhibited under possibility modals is only the tip of the Free Choice iceberg. The assumption that Free Choice can also, in some cases, strengthen unembedded disjunctions

to conjunctions has proven useful to account for properties of Warlpiri connectives *manu* (Bowler, 2014)[14] and children's conjunctive interpretation of *or* (Singh et al., 2016)[15].

Exhaustive participation and distributive implicature. With cumulative sentences, the analogy to Free Choice does not seem to hold. Consider (49). The basic meaning asserts that all nuts were cracked by a squirrel and the exhaustive participation inference we need to derive is one that asserts that all squirrels cracked a nut. This exhaustive participation inference certainly does not look like a Free Choice inference: an existential of the underlying meaning is indeed converted to a universal, but in addition an existing universal meaning in the underlying meaning is "*weakened*" to an existential.

(49) a. The squirrels ∃-cracked every nut.
b. **Attested meaning:**
$\forall x \in \mathsf{nut}, \exists y \prec \iota\mathsf{squirrels}, \mathsf{crack}(x)(y)$
$\forall y \prec \iota\mathsf{squirrels}, \exists y \in [\![\mathsf{nut}]\!], \mathsf{crack}(x)(y)$

At first blush, this inference is not familiar. To see more clearly, consider a case where there are only three squirrels - Scrat, Acorn and Waggs. Then, the underlying meaning derived by our proposal can be rewritten as in (50a) and the needed exhaustification inference as in (50b):

(50) a. $\forall x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{scrat}) \vee \mathsf{crack}(x)(\mathsf{acorn}) \vee \mathsf{crack}(x)(\mathsf{waggs})$
$\rightsquigarrow$ *every nut was cracked by Scrat, Acorn or Waggs*
b. $\exists x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{scrat})$
$\exists x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{acorn})$
$\exists x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{waggs})$

Observe how the meta-language paraphrase of (50a) (i.e. *every nut was cracked by Scrat, Acorn or Waggs*) does have the exhaustive participation inference in (50b). This is because the meta-language paraphrase has a *distributive implicature*. Distributive implicatures are attested when disjunctions are embedded under a quantifier (cf (51)), each disjunct is implicated to relate to at least one member of the domain of the quantifier.

(51) Every ambassador speaks Arabic, English or Mandarin.
$\rightsquigarrow$ at least one ambassador speaks Arabic.
$\rightsquigarrow$ at least one ambassador speaks English.
$\rightsquigarrow$ at least one ambassador speaks Mandarin.

The parallel does not stop at cumulative readings of *every*. Other quantifiers as well give rise to exhaustive participation inferences. For instance, *most*:

[14] The data of Szabolcsi and Haddican (2004) from Hungarian could also be interpreted as a *manu*-like connective, although the authors' interpretation is different.

[15] The conjunctive interpretation of "*or*" is not settled: some follow-up either replicate the pattern Tieu et al. (2015), while others attribute to confounds in design Skordos et al. (2020)

(52) a. The squirrels ∃-cracked most nuts.

b. $\mathsf{Most}\ x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{scrat}) \vee \mathsf{crack}(x)(\mathsf{acorn}) \vee \mathsf{crack}(x)(\mathsf{waggs})$
$\rightsquigarrow$ *most nuts were cracked by Scrat, Acorn or Waggs*

c. $\exists x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{scrat})$
$\exists x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{acorn})$
$\exists x \in \mathsf{nut}, \mathsf{crack}(x)(\mathsf{waggs})$

Correspondingly, these quantifiers give rise to distributive implicatures when they embed disjunctions[16]:

(53) Most ambassadors speak Arabic, English or Mandarin.
$\rightsquigarrow$ at least one ambassador speaks Arabic.
$\rightsquigarrow$ at least one ambassador speaks English.
$\rightsquigarrow$ at least one ambassador speaks Mandarin.

The parallel between exhaustive participation in cumulative sentences and distributive implicatures even extends to more than one quantifier, as happens in the video-game example example. Compare the inferences of the video-game example (54a) to that of a close paraphrase (55):

(54) **Cumulative sentences**

a. The video-games taught every quarterback ten new plays.

b. Every quarterback learned ten newplays from one of the video-games.
Video-game 1 taught one of the quarterbacks some play
Video-game 2 taught one of the quarterbacks some play
...

(55) **Distributive implicatures**[17]

a. Every quarterback learned ten new plays from video-game 1, 2 or 3

b. Every quarterback learned ten newplays from one of the video-games.
$\rightsquigarrow$ video-game 1 taught one of the quarterbacks some play
$\rightsquigarrow$ video-game 2 taught one of the quarterbacks some play
$\rightsquigarrow$ ...

Summarizing the points below, we see that each of the exhaustive participation inferences in our dataset have an implicature counterpart. We also find a split: the exhaustive participation inferences of non-cumulative sentences resemble Free Choice inferences whereas the exhaustive participation inferences of cumulative sentences resemble distributive implicatures.

[16] Credits for this observation goes to Filipe Hisao Kobayashi (p.c.)

[17] The video-game example is not the best example to demonstrate distributive implicatures of multiple quantifiers on, since the paraphrase in (56a) has all sorts of scope ambiguities. In addition, Denic (2020) notes that distributive implicatures compete with ignorance inferences when quantifiers have small domains. The following example is isomorphic to the one I used in the main text but is, I think, clearer:

(56) Every president gave at least ten ambassadors cash or judicial information.
$\rightsquigarrow$ at least one president gave at least one ambassador cash
$\rightsquigarrow$ at least one president gave at least one ambassador judicial information

Since it is not obvious how free choice inferences and distributive implicature form a natural class, the reader may justifiably be worried that we will be in need of two accounts of exhaustive participation: one for free-choice/non-cumulative sentences and one for distributive implicatures/cumulative sentences. However, it will not be so ; there will be a unified account of Free Choice and distributive implicatures. Before we can see this formally, I want to point out tentative evidence that natural languages do see a parallel between the two inferences.

As is well know, some items cross-linguistically lexicalize the behavior of Free Choice implicatures. Consider the German item *irgendein* (Kratzer and Shimoyama, 2017). Under a possibility modal, it gives rise to a free choice inference that any doctor is a permissible option:

(57) *Maria kann irgendeinen Arzt heiraten*
Mary must IRGENDEIN doctor marry

"*Mary can marry some doctor ; any doctor is allowed*"

However, the same item also triggers a "*Free Choice*" inference under a universal modal. In (58) too, it is implicated that any doctor is a permissible option.

(58) *Maria muss irgendeinen Arzt heiraten*
Mary must IRGENDEIN doctor marry

"*Mary must marry some doctor ; any doctor is allowed*"

However, note that this inference is nothing more than a distributive implicature in the modal realm. Indeed, imagining that there are only three doctors a, b and c, then the truth-conditions of (58) can be rendered logically as in (59). And these truth-conditions precisely match *mutatis mutandis* the distributive implicatures I reported on the ambassador example (51).

(59) $\forall w, \text{marry-a}(w) \vee \text{marry-b}(w) \vee \text{marry-c}(w)$
$\wedge \exists w, \text{marry-a}(w)$
$\wedge \exists w, \text{marry-b}(w)$
$\wedge \exists w, \text{marry-c}(w)$

The parallel does not go further than that, since *irgendein* does not trigger distributive implicatures with ordinary universal quantifiers like the German counterpart of *every*. Whatever the contribution of *irgendein* is seems to be bound to the presence of a modal quantifier. However, it does highlight a point of convergence between Free Choice and distributive implicatures which makes the unification that the account I will develop likely.

IV.2 Summary

The exhaustive participation inferences we are trying to derive not only match the signature of implicatures (polarity-sensitivity and context-sensitivity), they are formally identical to free choice in the case of non-cumulative sentences and distributive implicatures in the case of cumulative sentences.

Before we move on, I wish to point out the modularity of the analysis. With this identification, we could in principle stop our investigation here and claim that any account of the Free Choice and distributive implicatures would translate to an account of exhaustive participation. In some sense, the proposal is more general than whatever specific account of Free Choice and distributive implicatures I am now about to propose. However, there are specific claims about cumulativity that I will make (cf section VII.3), which will be impossible to state without a formal account of these inferences. This is why we genuinely need to spell out an analysis of FC/distributive implicatures.

V Free Choice and exhaustive participation in non-cumulative sentences

(A word of caution: the discussion will presuppose familiarity with Innocent Exclusion exhaustification and I refer the reader to Fox (2007) for a refresher)

In this section, I will review two accounts of free-choice: Fox (2007)'s recursive exhaustification and Bar-Lev and Fox (2017)'s innocent inclusion. Both accounts will satisfactorily translate to an account of exhaustive participation in *non*-cumulative sentences. Turning to distributive implicatures, I will review arguments by Crnič et al. (2015) why the standard account of these inferences is problematic ; their challenge will be shown to mirror a problem faced by Kratzer (2003) on cumulative sentences. Intriguingly, we will note that recursive exhaustification provides a satisfactory way out of Crnič et al. (2015)'s challenge, while the innocent inclusion approach seems to nail the coffin that Crnič et al. (2015) have prepared for us. All in all, because it provides a robust unified account of Free Choice and distributive implicatures, I will adopt the recursive approach.

V.1 Free Choice: recursive exhaustification and innocent inclusion.

Recursive exhaustification. Fox (2007) treats free choice inferences as arising from the recursive application of the innocent exclusion Exh operator. Specifically, he assumes the sentence (59) to have the structure in (60).

(60) Exh_2 Exh_1 you are allowed to eat apple or cake.

To guide intuitions about what this is meant to accomplish, we can reason along the following Gricean lines (inspired by Kratzer and Shimoyama (2017)): by leaving one of the disjuncts out, as in (61), the speaker would have conveyed that this disjunct was the only possibility. Since the speaker didn't leave either of the disjuncts out, then one concludes that neither disjunct is the only possibility. This means that both disjuncts are in fact possible, i.e. the free choice inference.

(61) a. You are allowed to eat apple. $\rightsquigarrow$ *you're allowed to eat apple and nothing else*
b. You are allowed to eat cake. $\rightsquigarrow$ *you're allowed to eat cake and nothing else*

This recursive reasoning is very powerful. Applying it to a simple disjunction such as (62), it would seem that the same reasoning would derive that you ate both apple and cake, by simply

negating (62a) and (62b) and their implicatures, just as in (61).

(62) Exh Exh You ate apple or cake.

a. You ate apple.
$\rightsquigarrow$ *You only ate apple.*

b. You ate cake.
$\rightsquigarrow$ *You only ate cake.*

However, these implicatures contradict the implicature that the hearer didn't eat both food items. The latter implication, a simple non-recursive implicature arising by competition with *and*, is derived by Exh_1, i.e. "*earlier*" in the computation. Since exhaustification in Fox (2007) is non-contradictory, this prevents the problematic recursive implicature that both disjuncts are true from being generated, since they are computed by Exh_2, i.e. "*later*". This correctly rules out any possibility of interpreting disjunction as conjunction.

Without the safeguard of the *not and* implicature, our disjunctions would, in positive environment, be strengthened to a conjunction. This type of strengthening, unwelcome for ordinary disjunctions, is exactly the result we wish to obtain for exhaustive participation inferences, replacing disjunctions with existentials and conjunctions with universals. Recall our non-cumulative sentence and its unstrengthened existential meaning:

(63) a. The dancers ∃-smiled.

b. **Unstrengthened meaning:**
$\exists x < \iota\mathsf{dancers},\ \mathsf{smiled}(x)$

c. **Attested meaning:**
$\exists x < \iota\mathsf{dancers},\ \mathsf{smiled}(x)$
$\forall x < \iota\mathsf{dancers},\ \mathsf{smiled}(x)$

To obtain this type of existential-to-universal strengthening on (63), we need two assumptions. First, to make a complete parallel between disjunctions and existentials, we need some counterpart to the individual disjuncts alternative of disjunction, seen in (62a) and (62b). Thinking of an existential as a grand disjunction, as in (64), these alternatives find a parallel in the *sub-domain alternatives* of the existential: alternatives where the existential is constrained to range over a smaller set of entities - in other words, these alternatives which have "*less disjuncts*". In our specific case, where the domain of the existential are the atomic parts of *the dancers*, the sub-domain alternatives are simply sentences where *the dancers* is replaced by a plurality of smaller size X, as in (64).

(64) a. $\exists x < \iota\mathsf{dancers},\ \mathsf{smiled}(x)$
$\leftrightsquigarrow \mathsf{smiled}(\mathsf{dancer\ 1}) \vee \mathsf{smiled}(\mathsf{dancer\ 2}) \vee \mathsf{smiled}(\mathsf{dancer\ 3}) \vee \ldots$

b. $\exists x < X, \mathsf{smiled}(x)$
where $X < \iota\mathsf{dancers}$

Second, to create the needed strengthening, we must prevent a implicature that *not all dancers smiled* from arising. Just as the *not and* implicature of disjunction blocks Free Choice strengthening of ordinary disjunction to conjunction, any *not all* implicature would block the Free Choice strengthening of existentials to universal. However, this *not all* implicature would

only arise if there were an ∀-*smiled* counterpart to ∃-*smiled* (mirroring fact that *and* is an alternative to *or*). We can reasonably assume, following Bar-Lev (2018b), that something like ∀-*smiled* doesn't exist ; we certainly have seen no data pointing to its existence.

All the assumptions are in place to derive the exhaustive participation inferences of our non-cumulative sentence (63). We apply recursive Exh to the sentence, assuming this Exh operator compares the prejacent to alternatives where the plural *the dancers* is replaced by smaller pluralities.

(65) $\textsc{Exh}_1$ $\textsc{Exh}_2$ [The dancers ∃-smiled.]$_\alpha$
Alts. to α:
$\{X$ ∃-smiled $\mid X \prec [\![\text{the dancers}]\!]\}$

As the reader can anticipate, the computation which delivers this result are somewhat involved. Existentials have more "*disjuncts*" than the simple disjunction seen for Free Choice. The complexity of the computations will only get worse, as we reach the cumulative sentences that this chapter builds towards. In an effort to properly separate the global narrative from the technical computations that it entails, I will perform the computation using the Python Exh package (Chatain, 2019) and informally explain why the computation delivers the result it does. The Exh package, available on PyPI, performs innocent exclusion exhaustification on user-defined formulas ; it can manage the type of recursive exhaustification we need to consider. The curious reader will be able follow along the relevant computations on the Jupyter notebook, following the [nb] links scattered in this chapter. In addition to streamlining the presentation, this method has the added advantage of minimizing the risk of errors and ensuring reproducibility.

By recursive exhaustification, (65) will be strengthened to have a universal meaning in the same way that Free Choice disjunction was strengthened [nb]. Namely, alternatives to the constituent that $\textsc{Exh}_2$ heads, which can be paraphrased as *among the dancers, only X smiled*, where *X* is strictly contained in *the dancers*, will all be negated by $\textsc{Exh}_1$. The added contribution of all of these negated inferences is that all the dancers smiled or none of them did. Together with the assertion of the prejacent, this entails that all the dancers did.

(66) a. **Alternatives to** $\textsc{Exh}_2$ [The dancers ∃-smiled.]$_\alpha$:
- Only Marie-Lou ∃-smiled.
- Only the dancers who are not Marie-Lou ∃-smiled.
- …

b. **Implicatures generated by $\textsc{Exh}_2$:**
1. Not only Marie-Lou ∃-smiled.
   ⇝ either Marie-Lou didn't smile or someone who wasn't Marie-Lou smiled.
2. Not only the dancers who aren't Marie-Lou ∃-smiled.
   ⇝ either Marie-Lou smiled or no one who wasn't Marie-Lou smiled.
   ⇝ either (Marie-Lou smiled and someone other than her did as well) or no one smiled*(together with 1)*
3. …

Innocent inclusion. Bar-Lev (2018b), rather than using the recursive Exh method of Fox (2007) for Free Choice as his account of the exhaustive participation inferences, uses instead

a new version of the Exh operator which performs all its duties in one fell swoop. His version of Exh starts by excluding innocently excludable alternatives, as the Exh of Fox (2007), but performs in addition an extra step of *inclusion*: it adds to the meaning of the prejacent whatever alternatives could not be excluded and can be included without contradiction. More formally, his recipe is outlined below:

(67) a. **Innocently excludable alternatives:**
$IE(\text{prejacent}, \text{Alts})$
$= \bigcap \left\{ C \subset \text{Alts} \;\middle|\; \begin{array}{l} C \text{ is a maximal subset of } \mathsf{Alts} \text{ s.t.} \\ \{\neg q \mid q \in C\} \cup \{p\} \text{ is consistent} \end{array} \right\}$

b. **Innocently includable alternatives:**
$II(\text{prejacent}, \text{Alts})$
$= \bigcap \left\{ C \subset \text{Alts} \;\middle|\; \begin{array}{l} C \text{ is a maximal subset of } \mathsf{Alts} \text{ s.t.} \\ \{\neg q \mid q \in C\} \cup \{\neg r \mid r \in IE(p, \mathsf{Alts})\} \cup \{p\} \text{ is consistent} \end{array} \right\}$

c. **Innocent inclusion exhaustification:**
$\text{Exh}_{II}(p, \mathsf{Alts}) = p \wedge \forall q \in IE(p, \mathsf{Alts}), \neg q \wedge \forall r \in IE(p, \mathsf{Alts}), \neg r$

Without changing our assumptions about the alternatives of (68), we can see the effect of *innocent inclusion* on our non-cumulative sentence [nb]. As was the case in the recursive Exh method, no alternative can be negated if Exh directly applies to constituent α. This is because the alternatives are *symmetric* ; for instance, negating the alternative that *some Bolivian dancers smiled* (cf (68b)) entails another alternative, namely that *some non-Bolivian dancers smiled*. Negating one of the two alternatives will prevent the other from being negated so neither can be *innocently* excluded.

(68) Exh_{II} [The dancers ∃-smiled.]$_\alpha$

a. **Alts. to α:** $\{X \text{ ∃-smiled} \mid X \prec [\![\text{the dancers}]\!]\}$
- The Bolivian dancers ∃-smiled. ()
- The non-Bolivian dancers ∃-smiled.
- …

b. **Predicted meaning:**
Some of the dancers smiled
and (via II) some Bolivian dancer smiled
and some non-Bolivian dancer smiled
and some dancer who is Marie-Lou smiled …
⇜⇝ *all dancers smiled*

However, all of these alternatives can be *in*cluded, as the new mechanism for Exh_{II} commands. By including (i.e. asserting) all of the alternatives in (68b) along with the prejacent, we correctly derive the exhaustive participation inference that *all dancers smiled*, as in (68b).

Summary Let us recap what we've seen so far. I presented two accounts of Free Choice effects *qua* implicatures. Both derive the exhaustive participation inference of non-cumulative sentences. The first account (Fox, 2007) relies on recursive exhaustification, by negating already exhaustified alternatives of the form *Only X smiled*. The second account (Bar-Lev and

Fox, 2017) relies on modifying the mechanism of exhaustification so that it would not simply exclude but also include.

VI The distributive implicatures: how Free Choice comes to the rescue

With these two accounts in the back of our mind, let us look at accounts of distributive implicatures. In our identification, these would correspond to exhaustive participation inferences in cumulative sentences like (69):

(69) a. The squirrels ∃-cracked every nut.
b. **Attested meaning:**
$\forall x \in \mathsf{nut}, \exists y \prec \iota\mathsf{squirrels}, \mathrm{crack}(x)(y)$
$\forall y \prec \iota\mathsf{squirrels}, \exists y \in [\![\mathrm{nut}]\!], \mathrm{crack}(x)(y)$

VI.1 The traditional account of distributive inferences and Crnič et al. (2015)'s challenge

In either a Gricean setting (Sauerland, 2004) or within the grammatical tradition, distributive implicatures are traditionally accounted for by negating the alternatives to the sentence obtained by leaving one disjunct out, as in (70a). The resulting inferences are not equivalent to the reported implicature that at least one ambassador speaks Arabic, at least one English, etc., but together with the prejacent, they entail it. Indeed, any ambassador who does not speak two out of the three languages must speak the third one, if the sentence is to be true. Since the implicatures in (70a) entail that there is such an ambassador for each pair of languages, we can find an ambassador who speaks any of the three languages, which is the desired distributive inference.

(70) Exh Every ambassador speaks Arabic, English or Mandarin. [nb]
a. **Negated alternatives:**
- not every ambassador speaks Arabic or English
  ⇝ *some ambassador doesn't speak Arabic or English*
- not every ambassador speaks Arabic or Mandarin
  ⇝ *some ambassador doesn't speak Arabic or Mandarin*
- not every ambassador speaks Mandarin or English
  ⇝ *some ambassador doesn't speak Mandarin or English*

As Crnič et al. (2015) point out, this *prima facie* satisfying result is problematic. They observe that the implicatures predicted by (70) are in fact stronger than the reported implicature in (51). Indeed, the implicatures of (70) not only imply the existence of an ambassador who speaks Arabic but of an ambassador who *only* speaks Arabic. However, they contend that the sentence[18] can be uttered if all ambassadors are, for instance, bilingual in two of the men-

[18] Or rather, one similar to it.

tioned languages, so long all languages are spoken by at least one ambassador[19]. In support of their contention, they offer experimental evidence that although speakers do reject sentences where one of the implicatures of (51) is not met, they in an overwhelming majority accept sentences where the implicatures drawn in (70b) are not met.

As it turns out, a problem isomorphic to Crnič et al. (2015)'s was independently encountered in discussion of cumulativity by Kratzer (2003). To see this, consider how one may translate the problematic account to the cumulative case of (71). Similarly to Free Choice, the counterpart of disjunction is the existentially interpreted verb ∃-*crack* ; the counterpart of the alternatives in (70) are alternatives where the subject plurality is replaced by sub-pluralities. Exhaustifying the sentence against these alternatives, we derive an inference that there is a nut that only Scrat cracked, one that only Acorn cracked, etc, which is *mutatis mutandis* the result obtained for (70). These inferences entail the exhaustive participation inferences ; for the strengthened meaning to be true, all three squirrels must have opened a nut. However, as with Crnič et al. (2015), they entail more than that.

(71) a. Exh Scrat, Acorn and Waggs ∃-cracked every nut. [nb]
b. **Alternatives:**
- Scrat and Acorn cracked every nut.
  ↔ *every nut was cracked by Scrat or Acorn*
- Scrat and Waggs cracked every nut.
- Waggs and Acorn cracked every nut.
- Waggs cracked every nut.
- etc.

On the particular example in (71), this is not problematic since every nut that was cracked by a squirrel must have been cracked by just one squirrel (a nut cannot be cracked twice!). However, (Kratzer, 2003, chap. 2, example 8) notes that this is in all generality a problematic inference to generate. Working in an event semantics, she derives the same inferences that we do, when she considers a semantic for *every* which incorporates a minimality operator[20]. She notes that these inferences would predict that (72) would be false in the scenario in (72b), since there isn't a mistake that either of the last two copy-editors caught alone, but the sentence is true in that scenario.

(72) a. The three copy-editors caught every mistake
b. **Strong Dist implicatures:**
for every copy-editor, there is a mistake that only they caught.
c. Two mistakes: Add and Omit
Copy-editor 1 caught Add
Copy-editor 2 caught Omit
Copy-editor 3 caught Omit

[19] And not every ambassador speaks all three languages. The latter constraint comes from the *and* implicature which I haven't shown.

[20] The correlation between the results of minimality and exhaustivity is most certainly not an accident. There are many similarities between her structures and ours: the scope of min with respect to the meta-language ∀ mirrors the scope of Exh and *every* above, the low-scoping existentials over events mirror our ∃-*smile*, etc. Chapter 3 will unravel these parallels in greater depth.

One way to temporarily evade Crnič et al. (2015)'s problem is to consider more alternatives. As Chemla and Spector (2011) note, we can avoid to generate the problematic implicatures if we consider that *every* has *some* as its scalar alternative[21], cf (73). These new alternatives create symmetry in the set of all alternatives. Negating "*some ambassador speaks Arabic or English*" prevents "*every ambassador speaks Mandarin*" from being negated and vice-versa. None of the alternatives can thus be innocently excluded [nb] and no scalar implicatures is generated. This is unfortunately only one half of a solution ; we not only need to explain why the problematic distributive implicatures are not generated but also how the attested distributive implicature are generated.

(73) a. Every ambassador speaks Arabic, English or Mandarin.
b. **Alternatives:**
- Every ambassador speaks Arabic or English.
- Every ambassador speaks Arabic or Mandarin.
- …
- Some ambassador speaks Arabic or English.
- Some ambassador speaks Arabic or Mandarin.
- …

A final note is that the problem uncovered by Crnič et al. (2015) runs deep. It also extends to other quantifiers than *every*, perhaps more blatantly. Distributive implicatures arise for other quantifiers than *every*[22]. As already discussed, distributive implicatures are also attested with other quantifiers than *every*. The inference derived by applying the exhaustification recipe in (70), replacing *every* with *many*, yields strong unattested inferences about the number of speakers of each language, i.e. (74b). Similarly here, we can avoid generating the inadequate implicatures by considering existential alternatives but we cannot generate the correct implicatures.

(74) a. Many ambassadors speak Arabic, English or Mandarin.
$\rightsquigarrow$ at least one ambassador speaks Arabic.
$\rightsquigarrow$ at least one ambassador speaks English.
$\rightsquigarrow$ at least one ambassador speaks Mandarin.
b. **Problematic inferences:**
not [many ambassadors speak Arabic or English]
not [many ambassadors speak Arabic or Mandarin]
not [many ambassadors speak English or Arabic]

The case of other quantifiers than *every* is especially relevant to us, since as we saw, exhaustive participation inferences do not just arise with *every*. We must make sure that our account extends to these other quantifiers as well.

Taming distributive implicatures with recursive exhaustification. The preceding discussion has established that tackling the problem of exhaustive participation inferences in cumulative sentences implies addressing Crnič et al. (2015)'s problem. To do so, let us observe

[21] This is a very reasonable assumption. This is traditionally invoked in discussions of indirect scalar implicatures, for instance the fact the *I didn't crack every nut* entails that *I cracked some*.

[22] Determining exactly which quantifiers yields that inference will be the subject of section VII.2.

that the distributive implicatures, repeated below in (75a), do not include negation in their paraphrases. This is surprising if these inferences are scalar implicatures derived by negating alternatives.

To go further, we could even construct what type of alternative would have to be negated in order to derive any one of the distributive implicatures: these would be alternatives like "*there isn't any ambassador who speaks Arabic*" or "*every ambassador does not speak Arabic*". Because these two sentences contain a negation not present in the sentence, it is delicate to explain how they could be alternatives to the sentence.

(75) a. Every ambassador speaks Arabic, English or Mandarin.
$\rightsquigarrow$ at least one ambassador speaks Arabic.
$\rightsquigarrow$ at least one ambassador speaks English.
$\rightsquigarrow$ at least one ambassador speaks Mandarin.

b. Many ambassadors speaks Arabic, English or Mandarin.
$\rightsquigarrow$ at least one ambassador speaks Arabic.
$\rightsquigarrow$ at least one ambassador speaks English.
$\rightsquigarrow$ at least one ambassador speaks Mandarin.

This reasoning, while certainly not unassailable, suggests that the distributive implicatures are not obtained by *ex*clusion, i.e. negating alternatives to the sentence, but by *in*clusion, i.e. asserting alternatives to the sentence. This is where our discussion of Free Choice makes an unexpected *re*appearance: when discussing Free Choice, we have seen two ways exhaustification can give rise to inclusion. One way was through recursive exhaustification ; the other through innocent inclusion. Can one of these proposals get us out of Crnič et al. (2015)'s challenge?

Innocent inclusion. Starting with innocent inclusion, recall that Bar-Lev and Fox (2017) posit that the semantics of Exh proceeds in two steps: first, Exh excludes alternatives, conforming to the traditional recipe ; second, it proceeds to include all of these alternatives it could not exclude.

Unfortunately, innocent inclusion will not solve the problem of Crnič et al. (2015). First, note that for Bar-Lev and Fox (2017)'s innocent inclusion approach to have a chance of success, we must make sure that the problematically strong scalar implicatures are not generated by the first step of exclusion. This is possible given Chemla and Spector (2011)'s remark that existential alternatives block these implicatures. With this assumption, no implicature is generated in exclusion step. The problem is that since no alternatives were excluded in the first step, they are all included in the second step, cf (76) [nb]. This generates an implicature that all ambassadors speak all three languages, which is even worse than than the original distributive implicature of Crnič et al. (2015)[23]

[23] One may think the problem could be avoided by considering the conjunctive alternatives of disjunctions. Since at least some of these conjunctive alternatives can be excluded, this prevents the inclusion step from including too much. Even if we leave aside the fact that this strategy doesn't apply to cumulative sentences (since ∃-cracked does not have conjunctive/universal alternatives), this fails to yield the distributive implicatures, cf [nb].

(76) a. Every ambassador speaks Arabic, English or Mandarin.

b. **Non-negatable ⇒ includable alternatives:**

- Every ambassador speaks Arabic or English.
- Every ambassador speaks Arabic.
- Every ambassador speaks Arabic or Mandarin.
- …
- Some ambassador speaks Arabic or English.
- Some ambassador speaks Arabic or Mandarin.
- …

c. **Predicted:**
Every ambassador speaks Arabic.
Every ambassador speaks English.
Every ambassador speaks Mandarin.

The failure of innocent inclusion to predict distributive implicatures will translate in a failure to account for exhaustive participation inferences in cumulative sentences. As we saw, these two inferences are *mutatis mutandis* the same.

Recursive exhaustification. This leaves us to consider how recursive exhaustification fares with respect to distributive implicatures. As the reader recalls, recursive exhaustification strengthens the sentence against already exhaustified alternatives, i.e. it negates not only alternatives but the implicatures that arise from them. This is achieved by the recursive Exh structure in (77a). Just as with innocent inclusion exhaustification, we must ensure that the problematic strong distributive implicatures of Crnič et al. (2015) are not generated by the first layer of Exh. Following Chemla and Spector (2011), this just means that we should consider the existential alternatives in addition to the universal alternatives.

(77) Exh Exh every ambassador speaks Arabic, English or Mandarin.

To understand the effect of the second layer of exhaustification exhaustification informally, I will capitalize on the Gricean intuition which I used in section V.1 to explain this account. Consider how hearers may interpret the alternative in (78) in a context where an informed speaker is answering the question of which language are represented among the ambassadors.

(78) *Of these three languages, which are spoken by the ambassadors?*
Every ambassador speaks Arabic or Mandarin.
⇝ no ambassador speaks English.

By uttering (79), the speaker seems to convey that the other language, English, is not spoken at all. A speaker who does not utter (79) would thus convey that the alternative and its implicatures are false, namely that either not every ambassador speaks Arabic or Mandarin or some ambassador speaks English. This simply means that some ambassador speaks English, which is the desired implicature.

The recursive exhaustification is a direct formal rendition of this Gricean intuition[nb]. The higher Exh will negate alternatives of the form in (79b), the counterpart of the alternatives that (78b) represents. Just as above, negating these inferences, together with the contribution

of the prejacent will result in the attested distributive implicatures: some ambassador speaks Arabic, some ambassador speaks English, some ambassador speaks Mandarin.

(79) a. Every ambassador speaks Arabic, English or Mandarin.
b. **Excludable alternatives:**
- Exh (Every ambassador speaks Arabic or Mandarin)
  $\leftrightarrow$ every ambassador speaks Arabic or Mandarin and it is not the case that some ambassador speaks English.
- Exh (Every ambassador speaks Arabic or English)
- Exh (Every ambassador speaks English or Mandarin)
- …

Note that in (79b), the implicature of *every ambassador speaks Arabic or Mandarin* is obtained by negating *some ambassador speaks English*. If this latter existential alternative had not been included in the set, the recursive exhaustification would have failed. TThis is one more reason for including existential alternatives in our set of alternatives.

VI.2 Summary

In section IV, we developed an analogy between exhaustive participation inferences in cumulative and non-cumulative sentences, on the one hand, and Free Choice and distributive implicatures on the other hand. I presented two accounts of Free Choice: innocent inclusion and recursive exhaustification.

For distributive implicature, the contribution of Crnič et al. (2015) was to show that distributive implicatures are not derivable by simply negating alternatives. Rather, distributive implicatures require the possibility to assert some alternatives. Innocent inclusion, while engineered for that result, asserted too many alternatives, aggravating the problem. By contrast, recursive exhaustification negates the already exhaustified alternatives, i.e. alternatives which already contain the negation of other alternatives. By double negation, this procedure ends up asserting exactly the inferences that we need.

Thus, recursive exhaustification is a unified account of distributive implicatures and Free Choice inferences. It is our best candidate for a unified account of exhaustification participation inferences. Let us recap the crucial assumptions made by this approach:

1. recursive Exh at root: exhaustifying already exhaustified alternatives..
2. existential alternatives to *every*
3. each sub-disjunction is an alternative to the embedded disjunction

VII Cumulative sentences

In the last section, we have seen that Free Choice inferences and distributive implicatures associated with disjunctions can receive a unified account through recursive exhaustification. As section IV has shown, these two inferences correspond to the exhaustive participation inferences of the non-cumulative and cumulative sentences of our dataset, so long as we are willing to see the existentials of verbal meanings (i.e. ∃-*smiled*) as grand disjunctions.

In the first part of this section, we apply the recursive exhaustification recipe to a simple non-cumulative and cumulative sentence. Although the packaging will look different, it is fundamentally the same analysis that we performed on scalar implicatures. The second part of this section is devoted to more intricate predictions. First, we will turn to other quantifiers than *every* ; we will try to spell out generalizations about which quantifiers give rise to exhaustive participation inferences and distributive implicatures. We will compare this with the prediction of recursive exhaustification. Second, we will then turn to an important topic which I have not mentioned so far: the asymmetries in cumulative readings of *every*. Fleshing out our assumptions somewhat more, we will see that these asymmetries follow from sensible restrictions on the placement of recursive exhaustification.

VII.1 Simple cases: non-cumulative sentences and cumulative readings of *every*

To port the results of Free Choice and distributive implicatures to exhaustive participation inferences, we must correspondingly adapt our assumptions. I assume that the sentences of our data-set are parsed with a recursive Exh mechanism at root and that each sub-plurality of a plural-referring expression is an alternative to this plural-referring expression expression. Our specific assumptions about the quantifier *every* are however retained. So our list of assumptions is given here. These are not final assumptions ; we will be making some addenda.

Assumptions

1. recursive Exh at root
2. each sub-plurality is an alternative to a plural-referring expression
3. existential alternatives to *every*

Non-cumulative sentences. With this set of assumptions, consider (80), whose counterpart in implicatures is the Free Choice sentence of (81).

(80) a. Exh Exh The dancers ∃-smiled [nb]

b. **Alternatives:**
- Exh (ML and W smiled.)
  = *only ML and W smiled*
- Exh (ML and B smiled.)
- Exh (W and B smiled.)
- Exh (ML smiled.)
- Exh (W and B smiled.)
- Exh (B smiled.)

c. **Predicted implicatures:**
⇝ Marie-Lou smiled
⇝ Wilfried smiled
⇝ Bianca smiled

If the set of dancers is Marie-Lou, Bianca and Wilfried, then the alternatives to (81a) are (81b). Negating all of these alternatives means that no sub-group of dancers are the only smilers.

This must mean that all dancers smiled, which is the exhaustive participation inference associated with non-cumulative sentences

(81) a. Exh Exh Marie-Lou or Wilfried or Bianca smiled
b. **Predicted implicatures:**
⇝ Marie-Lou smiled
⇝ Wilfried smiled
⇝ Bianca smiled

Cumulative readings of *every*. The same set of assumptions explains the exhaustive participation inferences of cumulative sentences with *every*. Consider (82), whose counterpart in implicatures is the sentence (83).

(82) a. Exh Exh The squirrels ∃-cracked every nut [nb]

b. **Alternatives:**
- Exh (S and W ∃-cracked every nut.)
  = *every nut was cracked by S or B*
  and no nut was cracked by A
- Exh (A and W cracked every nut.)
- Exh (S and A cracked every nut.)
- Exh (S cracked every nut.)
- Exh (A cracked every nut.)
- Exh (W cracked every nut.)

c. **Predicted implicatures:**
⇝ Scrat cracked a nut.
⇝ Acorn cracked a nut.
⇝ Waggs cracked a nut.

I assumed there are three squirrels: Scrat, Acorn and Waggs. Each of the alternative assert that some sub-group of squirrels contain all the nutcrackers. Negating all of them implies that all three squirrels are nutcrackers, which is the desired exhaustive participation inferences.

(83) a. Exh Exh Every nut was cracked by Scrat, Acorn or Waggs
b. **Predicted implicatures:**
⇝ Marie-Lou smiled
⇝ Wilfried smiled
⇝ Bianca smiled

This illustrates the core of the analysis of one non-cumulative sentence and the cumulative sentence with *every*.

VII.2 What about other quantifiers?

Upward-entailingness entails participation. So far, we considered a limited portion of our dataset. As the other elements of our dataset (repeated in (84)) attest, exhaustive participation inferences extend to many other quantifiers and even multiple quantifier:

(84) a. The ten squirrels cracked most nuts.
b. The ten squirrels cracked many nuts.
c. The ten video-games taught every quarterback two new plays.

As we discussed in section IV, these exhaustive participation inferences should mirror the distributive implicatures of the corresponding sentences:

(85) a. Most nuts were cracked by Scrat, Acorn or Waggs
b. Many nuts were cracked by Scrat, Acorn or Waggs
c. Every quarterback were taught two new plays by every video-game.

Because recursive exhaustification is so heavily dependent on what the alternatives and the prejacent are, it is not *prima facie* obvious that it would be able to explain the distributive implicatures and exhaustive participation inferences of other quantifiers. If the case of *every* is any clue, a necessary condition for deriving these inferences for other quantifiers will be that these quantifiers have an existential alternative, e.g. *some*. As we saw with *every*, the existential alternatives both blocks Crnič et al. (2015)'s problematic distributive implicatures and in the second round of exhaustification brings about the attested distributive implicatures.

As it turns out, one can prove a general mathematical guarantee that recursive exhaustification will derive the right result for most quantifiers, provided this condition and others are met:

The "*UE entails participation*" guarantee.

Let $\mathcal{Q}$ be a non-trivial quantifier, $\exists_C$ an existential quantifier with sub-domain alternatives.
If the following conditions hold:
- $\mathcal{Q}$ is upward-entailing,
- $\mathcal{Q}$ has *some* as its *only* alternative

then Exh Exh$(\mathcal{Q}x, \exists_C y, R(x, y))$ will be equivalent to the conjunction of the prejacent and the exhaustive participation inference that $\forall y, \exists x, R(x, y)$

The appendix (VII.4) gives the formal proof of the result. However, we can see why it holds on the particular case of *most*, as in (86).

(86) a. Exh Exh The squirrels ∃-cracked most nuts [nb]
b. **Alternatives:**
- Exh (Scrat and Waggs cracked most nuts)
  ≈ *most nuts were cracked by Scrat and Waggs and none by Acorn*
- Exh (Acorn and Waggs cracked most nuts)
- …

Consider what would happen if Acorn, one of the squirrels, didn't crack any nut. Then it would be true that most nuts were cracked by Scrat or Waggs, since the prejacent assert that most nuts were cracked by one of the squirrels and we know Acorn didn't contribute to the collective effort. That would make the first alternative of (87b) true. Because this alternative

is innocently excludable[24], we know that it can't possibly be true. By *reductio ad absurdum*, we show that Acorn must have cracked a nut. By symmetry, all squirrels must have cracked a nut.

Provided we ignore other alternatives these quantifiers might have, we then do predict that all of the sentences in (87) will have exhaustive participation inferences:

(87) a. The squirrels cracked most nuts
b. The squirrels cracked many nuts

Outside the guarantee's jurisdiction. The "*UE entails participation*" guarantee has clauses and it is natural to wonder what happens when one of these clauses is not met. For instance, what about exhaustive participation inferences with quantifiers which are not upward entailing?

For the case of "*no*", exhaustive participation inferences are not attested ; these inferences would in any case contradict the assertion[25]. But we can capture this case without the guarantee. As one can readily observe, the prejacent's meaning in (88b) is stronger than all of the alternatives and exclusion is impossible.

(88) a. The three squirrels cracked no nuts
$\not\rightsquigarrow$ *Acorn cracked a nut*
b. **Underlying meaning:**
$\neg\exists x \in \mathsf{nut}, \exists y \prec \mathsf{scrat} + \mathsf{acorn} + \mathsf{waggs}, \mathsf{cracked}(x)(y)$
c. **Alternatives:**
- Scrat and Waggs cracked no nuts
$\rightsquigarrow \neg\exists x \in \mathsf{nut}, \exists y \prec \mathsf{scrat} + \mathsf{waggs}, \mathsf{cracked}(x)(y)$
- …

For the case of downward-entailing quantifiers such as *less than 10*, the empirical picture is more difficult. I wasn't able to construct clear-cut data on whether *less than 10* does give rise to exhaustive participation inferences.

(89) Scrat and Acorn cracked less than 10 nuts.
$\overset{?}{\rightsquigarrow}$ *Scrat cracked a nut*

Correspondingly, whether these quantifiers do give rise to distributive implicatures is not clear:

(90) Less than 10 ambassadors speak Arabic, English or Mandarin.
$\overset{?}{\rightsquigarrow}$ *one ambassador speaks Arabic*

Another limitation of the "*UE entails participation*" guarantee is that it does not apply if any other alternatives are considered beyond the existential alternative of the quantifier $\mathcal{Q}$ and

[24] Part of the VII.4 is to prove that the alternative is indeed innocently excludable.
[25] As foreshadowed in section II.2, this explains why the "*exhaustive participation inference*" does not hold in generality.

the sub-domain alternatives. Yet, other alternatives sometimes have to be considered. Indeed cumulative sentences do give rise to scalar implicatures beyond the exhaustive participation inferences. For instance, (91) gives rise to the implicature that *not all* nuts were cracked:

(91) The ten squirrels cracked most of the nuts.
implicates: *not all of the nuts were cracked by a squirrel.*
implicates: *every squirrel cracked a nut.*

There is only so much one can prove with such generality so the lack of mathematical guarantees in such cases is to be expected. However, the computational tools of the package Exh (Chatain, 2019) provide reasons for optimism. We can for instance computationally determine what the predicted meaning of (91) is, with recursive exhaustification (cf (92)). It matches the attested reading of (91).

(92) a. Exh Exh The ten squirrels cracked most of the nuts. [nb]
b. **Computed result**:
$\mathsf{Most}\ x \in \mathsf{nut}, \exists y \in \mathsf{squirrel}, \mathsf{cracked}(x)(y)$
$\wedge \neg \forall x \in \mathsf{nut}, \exists y \in \mathsf{squirrel}, \mathsf{cracked}(x)(y)$ (*not all* implicature)
$\wedge \forall y \in \mathsf{squirrel}, \exists x \in \mathsf{nut}, \mathsf{cracked}(x)(y)$ (exhaustive participation)

Although this would need to be shown in all generality, everything proceeds in this case as if the generation of regular implicatures did not interfere with the generation of exhaustive participation inferences.

The final case to consider is what the theory predicts for the video-game example, repeated below in (93). In (93), two quantifiers are used, not just one. The mathematical guarantee "*UE entails participation*", which only considers the effect of one quantifier, is not applicable in such cases.

(93) The ten video-games taught every quarterback two new plays.
$\rightsquigarrow$ *every video-game taught some quarterback some play*

Where mathematical guarantees are unattainable, the numerical simulations of Exh come to the rescue. The result for a version of Schein's sentences are given below[26,27].

(94) a. Exh Exh The ten video-games taught every quarterback two new plays.
$\rightsquigarrow$ *every video-game taught some quarterback some play*
b. **Computed result** [nb]:
$\forall x \in \mathsf{quarterback}, \exists 2\ y \in \mathsf{plays}, \exists z \prec \iota\mathsf{video\text{-}games}, \mathsf{taught}(x)(y)(z)$
$\wedge \forall z \prec \iota\mathsf{video\text{-}games}, \exists x \in \mathsf{quarterback}, \exists y \in \mathsf{plays}, \mathsf{taught}(x)(y)(z)$
= *exhaustive participation inference*

[26] Computational complexity skyrockets in the video-game example so the simulation was performed on small domains.

[27] Here is a reason to see why this should hold. Consider how the pair *every video-game - two new plays* is formally equivalent to a quantifier over pairs. Seen in this light, this quantifier over pairs has existential entailments and has an existential over pair as an alternative. As a result, the "*existence entails participation*" guarantee and its strengthened version discussed in appendix VII.4 is almost applicable.

Summary and outlook. The results of this sub-section conclude our investigation of the dataset introduced in I. Before we move beyond it, let me recap the analysis so far. In section III, we introduced existential meanings into verbs, e.g. ∃-cracked, ∃-taught, etc. I showed how this simple assumption immediately predicted the truth-conditions of the negative version of the sentences of the dataset:

(95) a. The squirrels didn't crack every nut.
$\rightsquigarrow [\![\text{not [the squirrels } \exists\text{-crack every nut]}]\!] = \neg\forall x \in \mathsf{nut}, \exists y \prec \iota\mathsf{squirrels}, \mathsf{crack}(x)(y)$
b. The squirrels didn't crack most nuts.
c. The ten video-games didn't teach every quarterback two new plays.

Building on an analogy with Free Choice and Distributive Implicatures, I motivated the recursive exhaustification of Fox (2007) as a way to derive exhaustive participation inferences in positive environment. Recursive exhaustification only occurs in positive environment, preserving the prediction of (95). It computes strengthenings to the prejacent by comparing it with alternatives formed from the sentence by one of the following scalar substitutions: 1) substituting a quantifier with an existential, 2) substituting a plural-referring expression with a sub-plurality. Although the exact extent to which exhaustive participation inferences are derived could not be determined by procedure, numerical simulations show that these inferences are at least derived for all the sentences in the dataset.

(96) a. The squirrels cracked every nut.
$\rightsquigarrow [\![\text{Exh Exh the squirrels } \exists\text{-crack every nut}]\!]$
$= \forall x \in \mathsf{nut}, \exists y \prec \iota\mathsf{squirrels}, \mathsf{crack}(x)(y)$
$\wedge \forall y \prec \iota\mathsf{squirrels}, \exists x \in \mathsf{nut}, \mathsf{crack}(x)(y)$
b. The squirrels cracked most nuts.
c. The ten video-games taught every quarterback two new plays.

The following box summarizes the analysis:

Assumptions

1. verbs have existential meanings (e.g. ∃-*cracked*)
2. recursive Exh in positive environment
3. each sub-plurality is an alternative to a plural-referring expression
4. existential alternatives to quantifiers

VII.3 Asymmetries in cumulative readings

Asymmetries with *every*: the data We are now ready to move beyond the dataset. A famous property of cumulative sentences with *every* is that they exhibit subject/object asymmetries[28] (Champollion, 2010; Ferreira, 2005; Haslinger and Schmitt, 2018; Ivlieva, 2013; Kratzer, 2003). Consider (97a) and (97b): while (97a) has a sensible cumulative reading, (97b) seems to require every squirrel to have cracked every nut (a doubly-distributive reading)

[28] As far as I understand, Kratzer (2003) was the first to notice the pattern although cumulative reading of *every* were presented at least as early as Schein (1993).

(97) a. Every squirrel cracked the ten nuts. (#cumulative)
b. The ten squirrels cracked every nut. (✓ cumulative)

Although Kratzer (2003) initially described the asymmetry as an asymmery in thematic positions, Champollion (2010); Zweig (2008) shows that the asymmetry is one of c-command: a cumulative reading with *every* is only possible when *every* is c-commanded by a plural-referring expression. We can appreciate the truth of this fact by comparing all argument positions of a ditransitive predicate (examples from Chatain (2020)):

(98) Agent/Theme
a. The twelve challenges taught Hercules every cardinal virtue. (cumulative)
b. Every challenge taught Hercules the four cardinal virtues. (#cumulative)

(99) Agent/Goal
a. The ten servers sent every customer an e-mail. (cumulative)
b. Every server sent the ten customers an e-mail. (#cumulative)

(100) Goal/Theme
a. Anya gave the ten charities in Boxborough every penny she had earned.[29] (cumulative)
b. Anya gave every charity in Boxborough the fifteen checks she had earned. (#cumulative)

An important component of the mystery of the asymmetries is that movement operations do not license cumulative readings: if a cumulative reading was unavailable before movement, it remains unavailable after it, even if said movement has scopal effects. In Chatain (2020), I demonstrate this on *wh*-questions, passives and Russian scrambling.

Below, I repeat the evidence from Russian. Russian quantifier *kazhdyj* behaves like its English counterpart *every* ; it gives rise to cumulative reading in object position (101), not in subject position (102).

(101) **Cumulative reading is possible when *kazhdyj* is in object position**

Kazhdyj	*povar*	*otrkyl*	*ustricy*
every	cook.Nom.M.Sg	open.Perf	oyster.Pl.Acc

"*Every cook opened the oysters*[30]." (#background)

[30] Given that the object is not marked for definiteness in Russian, one may wonder why the Russian sentence cannot be read as "*every cook opened some oysters*". Under that reading, the sentence would not conflict with background knowledge. This is due to the presence of the perfective, which forces definite reading on bare plural objects (Krifka, 1992).

(102) **Cumulative reading is impossible when *kazhdyj* is in subject position**

Povara	*otrkyli*	*kazhduju*	*ustricu*
cook.Nom.Pl	open.Perf	every	oyster.Sg.F.Acc

"*The cooks opened every oyster*" (✓ background)

As (103) shows, scrambling the object DP above the subject DP in (102) does not give rise to a cumulative reading, even though that Russian scrambling makes the object c-commands the subject and usually has scopal effects (Antonyuk, 2006; Ionin, 2001; Stoops and Ionin, 2013).

(103)

Ustricy	*kazhdyj*	*povar*	*otrkyl*
oyster.Pl.Acc	every	cook.Nom.M.Sg	open.**Perf**

"*Every cook opened the oysters*" (#background)

Compiling the data from English and Russian, I form the following generalization:

Generalization

A cumulative reading between *every* and a plural-referring expression is only available when *every* is c-commanded by the plural-referring expression's base position.

Analysis. Let us first focus on the case of *every*. Within the theory of this chapter, the underlying meanings of sentences with *every* in subject position and the sentences with *every* in object position are parallel: the plural-referring expression, translated as an existential, takes scope under the universal quantifier.

(104) a. Every squirrel ∃-cracked the three nuts.
b. **Underlying meaning:**
$\forall y \in \mathsf{squirrel}, \exists x < \iota\mathsf{nuts}, \mathsf{crack}(x)(y)$

(105) a. The three squirrels ∃-cracked every nut.
b. **Underlying meaning:**
$\forall y \in \mathsf{nut}, \exists x < \iota\mathsf{squirrels}, \mathsf{crack}(y)(x)$

If the goal is to find asymmetries, such a parallel in truth-conditions is worrying. But recall that the underlying meaning is only ever seen in negative environments. This means that we expect the truth-conditions of the negation of (104) and (105) to behave in a parallel manner. This prediction is borne out: as reported in Križ and Chemla (2015), the negation of (104) has the truth-conditions in (106). These truth-conditions mirror[31] the truth-conditions of the negation of (105), which we already discussed in section III.

[31] Interestingly, these sentences seem to differ in their implicatures. While (106a) implicates that some squirrel cracked the three nuts, (106b) implicates that some squirrels cracked some nuts. If the treatment of exhaustive participation inferences that I propose is correct, these implicatures would parallel the strengthened indirect implicature of (106):

(107) a. Not every squirrel cracked the three nuts.
=*not every squirrel cracked a nut*
=*some squirrel cracked no nut*

b. The three squirrels didn't crack every nut.
=*not every nut was cracked by a squirrel*
=*some nut wasn't cracked by any squirrel*

The parallel in underlying meanings suggest that any difference between subject *every* sentences and object *every* sentences is due to the way the two sentences are strengthened in positive environments.

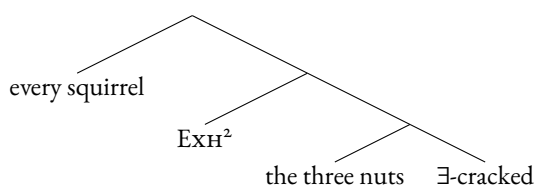
Problematically, using the recursive exhaustification at root that we have been using so far, as in (108), is bound to deliver the same strengthening for both sentences. (I will use the symbol Exh² for recursive exhaustification). Indeed, these sentences have the same underlying meaning and identical alternatives. Both sentences, as it stands, will receive a cumulative reading, contrary to fact.

(108) a. Exh² Every squirrel ∃-cracked the three nuts.
⇝ *every squirrel cracked a nut and every nut was cracked by a squirrel*.

b. Exh² The three squirrels ∃-cracked every nut.
⇝ *every squirrel cracked a nut and every nut was cracked by a squirrel*

The reason for the asymmetry stems, I contend, from the scope of exhaustivity. So far, I have assumed that all exhaustification happens at root. Consider what would happen for different placements of Exh². When *every* is in subject position, Exh² can be inserted in the scope of *every*, while still c-commanding *the three nuts*, as shown in (109).

(109)



In this position, Exh² applies directly to the existential over parts of the nuts. As we saw in section V, this is precisely the configuration in which a Free Choice-like inference is generated. Concretely, this means that *the three nuts* is strengthened to a universal ; the sentence receives a doubly-distributive reading[nb].

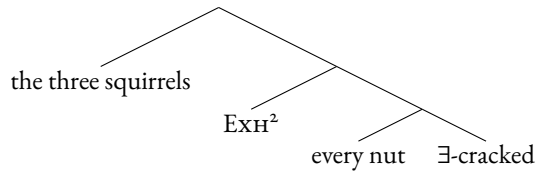
(110) $[\![(109a)]\!] = \forall y \in \mathsf{squirrel},\ \text{Exh}^2(\exists x \prec \mathsf{nut}_1 + \mathsf{nut}_2 + \mathsf{nut}_3,\ \mathsf{cracked}(x)(y))$
$\textbf{alts: } \exists y \prec \mathsf{nut}_1 + \mathsf{nut}_2,\ \mathsf{cracked}(x)(y)), \ldots$
$= \forall x \in \mathsf{squirrel}, \forall y \prec \mathsf{nut}_1 + \mathsf{nut}_2 + \mathsf{nut}_3,\ \mathsf{cracked}(x)(y)$

By contrast, when *every* is in a object position, there is no position where Exh² can be placed in which it both c-commands the plural and is c-commanded by *every*. There are embedded

(106) I didn't show every boy some of my paintings. ⇝ *I showed every boy some but not all of my paintings.*
To my knowledge, these types of implicatures are not accounted for, nor discussed by previous literature.

position but they fail to c-command the plural-referring expressions *the three squirrels*. Failing to c-command *the three squirrels* means that the alternatives which Exh compares will not contain any sub-domain alternatives ; no cumulative strengthening is derived. This is illustrated in (111).

(111) a.



b. $[\![(110a)]\!] = \iota\mathsf{squirrels}\ \lambda y.\ \text{Exh}^2(\forall x \in \mathsf{nuts},\ \mathsf{cracked}(x)(y))$
$\textbf{alts: } \exists y \in [\![\text{nut}]\!],\ \mathsf{cracked}(x)(y))$
$= \iota\mathsf{squirrels}\ \lambda y.\ \forall x \in \mathsf{nuts},\ \mathsf{cracked}(x)(y)$

The following chart summarizes our discussion so far:

	subject *every*	object *every*
root Exh²	cumulative	cumulative
embedded Exh²	doubly-distributive	vacuous

From this chart, I make the following simple proposal: Exh² must apply at all positions (Magri, 2011). This type of structure is summarized in (112) below. With object *every*, this assumption is innocuous because embedded Exh² does not result in strengthening. The composition proceeds as if there was only a root Exh²; as we saw, this is how the cumulative reading is generated. With subject *every*, embedded Exh² results in the attested doubly-distributive reading. The root Exh² cannot strengthen the meaning beyond further.

(112) a. Exh² the three squirrels Exh² ∃-cracked every nut. [nb]
b. Exh² every squirrel Exh² ∃-cracked the three nuts. [nb]

In short, the asymmetry between the cumulative reading of object *every* and the doubly-distributive reading of subject *every* is that only in the latter case, there is a position below the quantifier where the sub-domain alternatives are visible and strengthening can happen. Assuming that strengthening not only can but must happen, through the postulate that exhaustification applies in all positions, creates the split between subject and object *every* in positive environment.

The assumption that recursive exhaustification must apply in all positions is the last assumption that I will make in this chapter about cumulativity. Adding it to the list of assumptions, we arrive at the following final theory:

Assumptions

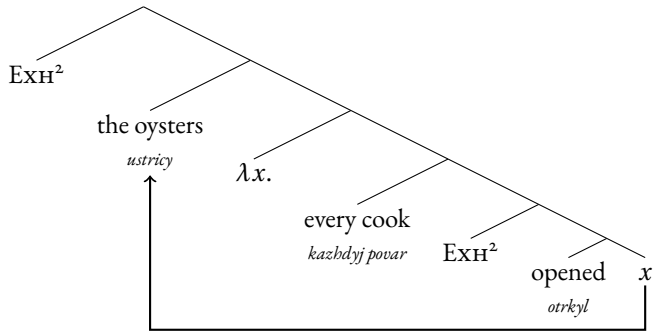
1. verbs have existential meanings (e.g. ∃-*cracked*).
2. recursive Exh in positive environment *in all positions.*.
3. each sub-plurality is an alternative to a plural-referring expression.
4. existential alternatives to quantifiers.

In the last part of the section, I turn to the other two facts about asymmetries that I outlined in the introduction: 1) that movement does not create new cumulative readings, 2) that quantifiers over pluralities do not give rise to asymmetries (cumulative readings available from both positions) - such as partitive quantifiers - but quantifiers over singularities do - non-partitive quantifiers -.

The effect of *movement*. Descriptively, cumulative readings will arise when *every* is c-commanded by the base position of a plural-referring expression and a doubly-distributive reading will otherwise arise. This analysis developed above allows for a more precise statement: as we discussed earlier, a doubly-distributive reading will arise whenever there is a position in the scope of the quantifier where sub-domain alternatives of the plural-referring expression are visible to Exh^2 ; otherwise, a cumulative reading is generated.

This refined generalization helps us understand why movement would not create new cumulative readings. Consider the structure of (113) for the Russian sentence corresponding to *every cook opened the oysters*, where the object scrambles above the subject.

(113)



While the movement does alter the c-command hierarchy, it leaves a trace in its base position. This means that despite movement, there is still a position within the scope of the quantifier where sub-domain alternatives - the sub-domain alternatives to the trace - can lead to strengthening. The doubly-distributive reading is still generated.

VII.4 Ordinary cumulative readings

At last, we are ready to tackle ordinary cumulative sentences, like (114). Ordinary cumulative sentences do not raise particular issues for the theory, but the full set of assumptions made in this chapter are necessary to account for it.

(114) a. The ten squirrels cracked the fifteen nuts.
b. The ten squirrels didn't crack the fifteen nuts.

The negative case in (114b) is the simplest. In the scope of negation, no strengthening through Exh^2 occurs ; both plural-referring expressions are interpreted as existential over parts of their denotation in the scope of negation. The predicted meaning matches the attested meaning: *no squirrel cracked any nut*.

(115) a. not [the ten squirrels ∃-cracked the fifteen nuts]

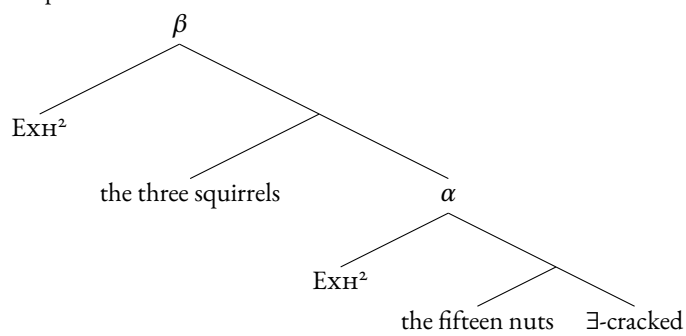
b. $[\![(115a)]\!] = \neg\exists x \prec \iota\mathsf{squirrels}, \exists y \prec \iota\mathsf{nuts},\ \mathsf{cracked}(y)(x)$

$\leftrightsquigarrow$ *no squirrel cracked any nut*

In the positive case of (116a), exhaustification is active. As seen in the last section, we need to include one Exh² in all positions.

(116) a. The ten squirrels cracked the fifteen nuts.

b.



c.

The computation is arduous [nb], but we can develop a simple intuition for how it will run. In the embedded position α, Exh² operates over the sub-domain alternatives of the existential represented by *the nuts*, cf (117). Because there is no intervening quantifier, *the nuts* will be strengthened to a universal (i.e. a Free Choice-like inference), cf (117a).

(117) $[\![\alpha]\!] = \lambda X.\ \text{Exh}^2(\exists y \prec \iota\mathsf{nuts}, \exists x \prec X,\ \mathsf{cracked}(y)(x))$

alts: $\exists y \in \mathsf{nut}_1 + \mathsf{nut}_2, \exists x \prec X\ \mathsf{cracked}(y)(x)), \ldots$

$= \forall y \prec \iota\mathsf{nuts}, \exists x \prec X,\ \mathsf{cracked}(y)(x)$

$\leftrightarrow$ *every nut was cracked by one of X*

In the root position β, Exh² operates over the sub-domain alternatives of *the squirrels*[32]. Here however, the existential represented by *the squirrels* finds itself in the scope of the universal corresponding to *the nuts* which was created by the first strengthening. The situation is entirely parallel to the case of cumulative readings of *every* ; Exh² will generate a distributive-like implicature. Together with the prejacent, this will create the cumulative reading.

(118) a. $[\![\beta]\!] = \text{Exh}^2(\forall y \prec \iota\mathsf{nuts}, \exists x \prec \iota\mathsf{squirrels}\ \mathsf{cracked}(y)(x))$

alts: $\exists y \in [\![\mathsf{nut}]\!], \exists x \prec \iota\mathsf{scrat} + \mathsf{waggs}\ \mathsf{cracked}(y)(x)), \ldots$

$= \forall y \prec \iota\mathsf{nuts}, \exists x \prec \iota\mathsf{squirrels},\ \mathsf{cracked}(x)(y)$

$\wedge \forall x \prec \iota\mathsf{squirrels}, \exists y \prec \iota\mathsf{nuts},\ \mathsf{cracked}(y)(x)$

$\leftrightarrow$ *cumulative reading*

All in all, the computation raises no particular issue. The object is first strengthened to a universal meaning ; from then on, the situation is entirely parallel to the case of cumulative readings of *every*.

[32] In addition to the sub-domain alternatives of *the nuts*.

Conclusion and further puzzles.

This chapter laid the foundations of a theory of cumulativity. This theory of cumulativity is special in that it does justice to the homogeneity properties of cumulative readings and gives an account of the truth-conditions of negative sentences, which have been ignored by previous approaches (cf chapter 3). More than that, I hope to have shown that two generalizations about the meaning of cumulative sentences can only be discovered by carefully comparing cumulative sentences in positive and negative environments. The first generalization claims that the part of the meaning which is common to both positive and negative environments is obtained by translating all plural-referring expressions by low-scope existentials. The second generalization asserts that the part of the meaning which only occurs in positive environments asserts all individuals in the plurals exhaustively took part in the action described by the verb.

The theory, whose assumptions are repeated in the box below, builds at its core existential quantification over parts in the meaning of the verb. With this assumption alone, the negative versions of all sentences in (118) are all covered. Then, I proceeded to identify the exhaustive participation inferences of positive sentences with Free Choice/distributive implicatures, which independently require an account. I argued that recursive exhaustification provides such an account. While the prediction of this account may sometimes be intractable, I showed that it delivered correct results as far as our dataset is concerned. Furthermore, it provided an understanding of the asymmetries in cumulative reading.

Assumptions

1. verbs have existential meanings (e.g. $\exists$-*cracked*).
2. recursive Exh in positive environment *in all positions.*.
3. each sub-plurality is an alternative to a plural-referring expression.
4. existential alternatives to quantifiers.

There are some gaps in the theory of this chapter which later chapters will fill. First, this chapter has unabashedly ignored the possibility of collective action. The sentence (119a) is still true if for every nut, it took multiple squirrels to crack it, each squirrel gnawing at one side of the nut, and all squirrels participated in one cracking. In this context however, the predicted (119b) would wrongly come out false since for no nut n can we find a squirrel s of which it is true that s cracked n. In chapter 2, I remedy this deficit of the theory.

(119) a. The squirrels cracked the nuts

b. $\forall x \prec \iota\mathsf{squirrels}, \exists y \prec \iota\mathsf{nuts}, \mathsf{cracked}(y)(x)$
$\wedge \forall y \prec \iota\mathsf{nuts}, \exists x \prec \iota\mathsf{squirrels}, \mathsf{cracked}(y)(x)$

Another limitation of the theory is that it currently does not predict the possibility of phrasal distributivity. For instance, (120) is only predicted to mean (120a), whereas it can also mean (120b).

(120) The squirrels cracked at least 4 nuts.

a. **Collective/Cumulative:**

the number of nuts cracked by any squirrels exceeds 4 and all squirrel cracked some nuts

b. **Distributive:**
for every squirrel, that squirrel cracked more than 3 nuts

From the perspective of the theory, those readings could be achieved if, contrary to my generalizations, the existentials associated with the interpretation of plural-referring expressions could take a higher scope. For instance, if we identified the operator of distributivity with the existential interpretation of plurals and allowed the latter to have free-scope, as in Bar-Lev (2018b)'s theory.

But I find reasons to distinguish the existential meanings needed for cumulative readings from those needed for distributive readings. As Champollion (2016a,b) argues, phrasal distributivity is typically atomic, unless context makes a cover available, whereas cumulative reading are typically non-atomic without need for contextual support. Furthermore, items which do not give rise to phrasal distributivity may give rise to cumulative readings (de Vries, 2015), suggesting that distributivity needs a theory of its own. It is the role of chapter 4 to develop this theory.

References

Antonyuk, S. (2006). The scope of quantifier phrases in Russian: A QR analysis. *Linguistics in the Big Apple*.

Bar-Lev, M. E. (2018a). *Free Choice, Homogeneity, and Innocent Inclusion*. PhD thesis.

Bar-Lev, M. E. (2018b). Homogeneity as an implicature.

Bar-Lev, M. E. and Fox, D. (2017). Universal Free Choice and Innocent Inclusion. In *Semantics and Linguistic Theory*, volume 27, pages 95–115.

Bassi, I. and Bar-Lev, M. E. (2018). A unified existential semantics for bare conditionals. In *Proceedings of Sinn Und Bedeutung*, volume 21, pages 125–142.

Bayer, S. L. (2013). *Confessions of a Lapsed Neo-Davidsonian: Events and Arguments in Compositional Semantics*. Routledge.

Beck, S. and Sauerland, U. (2000). Cumulation is needed: A reply to Winter (2000). *Natural language semantics*, 8(4):349–371.

Bowler, M. (2014). Conjunction and disjunction in a language without'and'. In *Semantics and Linguistic Theory*, volume 24, pages 137–155.

Buccola, B. and Spector, B. (2016). Modified numerals and maximality. *Linguistics and Philosophy*, 39(3):151–199.

Champollion, L. (2010). Cumulative readings of every do not provide evidence for events and thematic roles. In *Logic, Language and Meaning*, pages 213–222. Springer.

Champollion, L. (2016a). Covert distributivity in algebraic event semantics. *Semantics and Pragmatics*, 9:11–15.

Champollion, L. (2016b). Overt distributivity in algebraic event semantics. *Semantics and Pragmatics*, 9(16):1–65.

Chatain, K. (2019). Exh: A Python package to compute innocent exclusion/inclusions exhaustivity.

Chatain, K. (2020). Cumulative readings of "every" and leaks. In *Proceedings of SALT 30*.

Chemla, E. and Spector, B. (2011). Experimental evidence for embedded scalar implicatures. *Journal of semantics*, 28(3):359–400.

Crnič, L., Chemla, E., and Fox, D. (2015). Scalar implicatures of embedded disjunction. *Natural Language Semantics*, 23(4):271–305.

de Vries, H. (2015). *Shifting Sets, Hidden Atoms: The Semantics of Distributivity, Plurality and Animacy*. PhD Thesis, Utrecht University.

Denic, M. (2020). Probabilistic informativeness in implicature computation: The case of embedded disjunctions.

Ferreira, M. M. B. (2005). Event quantification and plurality.

Fox, D. (2007). Free choice and the theory of scalar implicatures. In *Presupposition and Implicature in Compositional Semantics*, pages 71–120. Springer.

Haslinger, N. and Schmitt, V. (2018). Scope-related cumulativity asymmetries and cumulative composition. In *SALT 28*.

Ionin, T. (2001). The one girl who was kissed by every boy: Scope, scrambling and discourse function in Russian. *Proceedings of ConSole X*, pages 65–80.

Ivlieva, N. (2013). *Scalar Implicatures and the Grammar of Plurality and Disjunction*. PhD thesis, Massachusetts Institute of Technology.

Kratzer, A. (2000). Schein's argument. In *The Event Argument*.

Kratzer, A. (2003). The event argument and the semantics of verbs. In *The Event Argument and the Semantics of Verbs*. ms edition.

Kratzer, A. and Shimoyama, J. (2017). Indeterminate pronouns: The view from Japanese. In *Contrastiveness in Information Structure, Alternatives and Scalar Implicatures*, pages 123–143. Springer.

Krifka, M. (1992). Thematic relations as links between nominal reference and temporal constitution. In Sag, I. A. and Szabolcsi, A., editors, *Lexical Matters*, pages 29–53. CSLI Publications, Chicago University Press.

Kriz, M. (2015). Aspects of homogeneity in the semantics of natural language.

Križ, M. (2016). Homogeneity, non-maximality, and all. *Journal of Semantics*, 33(3):493–539.

Kriz, M., Brochhagen, T., Roelofsen, F., and Theiler, N. (2015). Homogeneity, trivalence, and embedded questions. In *Proceedings of the 20th Amsterdam Colloquium*, pages 207–16.

Križ, M. and Chemla, E. (2015). Two methods to find truth-value gaps and their application to the projection problem of homogeneity. *Natural Language Semantics*, 23(3):205–248.

Križ, M. and Spector, B. (2017). Interpreting Plural Predication : Homogeneity and Non-Maximality. pages 1–53.

Lasersohn, P. (1999). Pragmatic halos. *Language*, pages 522–551.

Löbner, S. (2000). Polarity in natural language: Predication, quantification and negation in particular and characterizing sentences. *Linguistics and Philosophy*, 23(3):213–308.

Magri, G. (2011). Another argument for embedded scalar implicatures based on oddness in downward entailing environments. *Semantics and Pragmatics*, 4:6–1.

Magri, G. (2014). An Account for the Homogeneity Effect Triggered by Plural Definites and Conjunction Based on Double Strengthening. In *Pragmatics, Semantics and the Case of Scalar Implicatures*, pages 99–145. Springer.

Malamud, S. (2012). The meaning of plural definites: A decision-theoretic approach. *Semantics and Pragmatics*, 5:1–3.

Roberts, C. (1987). *Modal Subordination, Anaphora, and Distributivity*. PhD thesis.

Sauerland, U. (2004). Scalar implicatures in complex sentences. *Linguistics and philosophy*, 27(3):367–391.

Scha, R. (1984). Distributive, collective and cumulative quantification. In *Truth, Interpretation and Information; Selected Papers from the 3rd Amsterdam Colloquium, Dordrecht, Holland*, pages 131–158.

Schein, B. (1993). *Plurals and Events*, volume 23. Mit Press.

Schwarzschild, R. (1993). Plurals, presuppositions and the sources of distributivity. *Natural Language Semantics*, 2(3):201–248.

Singh, R., Wexler, K., Astle-Rahim, A., Kamawar, D., and Fox, D. (2016). Children interpret disjunction as conjunction: Consequences for theories of implicature and child development. *Natural Language Semantics*, 24(4):305–352.

Skordos, D., Feiman, R., Bale, A., and Barner, D. (2020). Do Children Interpret 'or' Conjunctively? *Journal of Semantics*, 37(2):247–267.

Sternefeld, W. (1998). Reciprocity and cumulative predication. *Natural language semantics*, 6(3):303–337.

Stoops, A. and Ionin, T. (2013). Quantifier scope and scrambling in Russian: An experimental study. In *Proceedings of the Annual Workshop on Formal Approaches to Slavic Linguistics: The Bloomington Meeting*.

Szabolcsi, A. and Haddican, B. (2004). Conjunction Meets Negation: A Study in Cross-linguistic Variation. *Journal of Semantics*, 21(3):219–249.

Tieu, L., Romoli, J., Zhou, P., and Crain, S. (2015). Children's knowledge of free choice inferences and scalar implicatures. *Journal of Semantics*, 33(2):269–298.

Zobel, S. (2016). On the (in) definiteness of impersonal pronouns. *Linguistica*, 56(1):363–374.

Zweig, E. (2008). *Dependent Plurals and Plural Meaning*. PhD thesis.

Appendix

VII.5 Formal results on recursive exhaustification

We set out to show the following fact in all generality:

The "*UE entails participation*" guarantee.

Let $\mathcal{Q}$ be a non-trivial quantifier, $\exists_C$ an existential quantifier with sub-domain alternatives.
If the following conditions hold:
- $\mathcal{Q}$ is upward-entailing,
- $\mathcal{Q}$ has *some* as its *only* alternative

then $\textsc{Exh}\,\textsc{Exh}(\mathcal{Q}x, \exists_C y, R(x,y))$ will be equivalent to the conjunction of the prejacent and the exhaustive participation inference that $\forall y, \exists x, R(x,y)$

Here are some additional assumptions about quantifiers which are reasonable to make:

- **permutation invariance+conservativity:** the truth $\mathcal{Q}x \in D, P(x)$ only depends on $P \cap D$'s cardinality[33]. Call `quant` the set of cardinalities $P \cap D$ where $\mathcal{Q}x \in D, P(x)$ is true. Because $\mathcal{Q}$ has existential entailments, `quant` is non-empty and contains one number between 1 and the cardinality of D.

I write D for the full domain of the existential quantifier, D_1 the domain of $\mathcal{Q}$, C for any strict non-empty sub-domain of D, $\bar{C}$ for the complementary of C in D (which is also non-empty because C is a strict subset). I write $\textsc{Exh}_1$ for the higher exhaustification operator and $\textsc{Exh}_2$ for the lower exhaustification operator.

By our assumptions, the alternatives that $\textsc{Exh}_1$ operates on are of one of two types:

[33] The more general statement is that the truth of the quantified statement depends on the cardinalities of $P \cap D$ and D. Since we're holding D fixed in our reasoning, I'm dropping the latter cardinality

(121) a. $\textsc{Exh}\mathcal{Q}x, \exists y \in C, R(x, y)$
b. $\textsc{Exh}\exists x, \exists y \in C, R(x, y)$

Call these alternatives the *higher* alternatives Let's gain some insights into what the *higher* alternatives mean. Each of them is the exhaustification of a prejacent against the set of alternatives (the same for all formulas in (121)), let's call them the *lower* alternatives, which are the following:

(122) a. $\mathcal{Q}x, \exists y \in C, R(x, y)$
b. $\exists x, \exists y \in C, R(x, y)$

Similarly, we define the *higher* prejacent and the *lower* prejacent.

We can completely determine the meaning of one type of *higher* alternative:

(123) **Fact (a).** $\textsc{Exh}(\exists x, \exists_C y, R(x, y))$ is equivalent to $\exists x, \exists_C y, R(x, y) \wedge \neg\exists x, \exists_{\bar{C}} y, R(x, y)$

Let us distinguish two cases. To be brief

- $\mathcal{Q}$ is equivalent to $\exists$. In other words, $\mathsf{quant} = \{1, \ldots, \#D\}$. In that case, all alternatives of the form $\mathcal{Q}x, \exists y \in C_0, R(x, y)$ are equivalent to $\exists x, \exists y \in C_0, R(x, y)$ and we can disregard them. In that case:

Maximal consistent set:	for some z in C, $\{\exists x, \exists y \in C_0, R(x, y) \mid z \notin C_0\}$ consistent for worlds where $R = D \times \{z\}$ maximal because $\neg\exists x, \exists y \in C_0, R(x, y)$ and $\neg\exists x, \exists y \in C_1, R(x, y)$ contradict the prejacent when $C_0 \cup C_1 = C$
IE alternatives:	$\{\exists x, \exists y \in C_0, R(x, y) \mid C_0 \cap C = \varnothing\}$

- $\mathcal{Q}$ is not equivalent to $\exists$. Then, there must be some cardinality n greater than 0 that is not in quant ; call S any subset of D_1 of that cardinality. In that case:

Maximal consistent set:	for some z in C, $\{\exists x, \exists y \in C_0, R(x, y) \mid z \notin C_0\} \cup \{\mathcal{Q}x, \exists y \in C_0, R(x, y) \mid \forall C_0\}$ consistent for worlds where $R = S \times \{z\}$ maximal because $\neg\exists x, \exists y \in C_0, R(x, y)$ and $\neg\exists x, \exists y \in C_1, R(x, y)$ contradict the prejacent when $C_0 \cup C_1 = C$
IE alternatives:	$\{\exists x, \exists y \in C_0, R(x, y) \mid C_0 \cap C = \varnothing\} \cup \{\mathcal{Q}x, \exists y \in C_0, R(x, y) \mid \forall C_0\}$

To prove this, I'll resort I claim that the minimal worlds are the worlds picked in which R (restricted to $D_1 \times D$) is only true of pairs of $D_1 \times C$. To see this, take one such world w and consider the pair (x, y) true of R. This world makes the following alternative true:

- $\exists x, \exists y \in C, R(x, y)$, where C contains y

(124) **Fact (b).** $\textsc{Exh}(\mathcal{Q}x, \exists y \in C, R(x, y))$ is stronger than $\textsc{Exh}(\exists x, \exists y \in C, R(x, y))$

By fact (a), $P := \textsc{Exh}(\exists x, \exists y \in C, R(x,y))$ is equivalent to $\exists x, \exists_C y, R(x,y) \wedge \neg \exists x, \exists_{\bar{C}} y, R(x,y)$ By definition of exhaustification, $Q := \textsc{Exh}(\mathcal{Q} x, \exists y \in C, R(x,y))$ will be equivalent to $\mathcal{Q} x, \exists y \in C, R(x,y) \wedge \neg a_1 \wedge \neg a_2 \ldots$. Since $\mathcal{Q}$ entails existence, we can already see Q entails the prejacent of P, i.e. $\exists x, \exists y \in C, R(x,y)$, which is one half of the meaning of P. We now just need to show that Q entails the other half, namely $\neg \exists x, \exists y \in \bar{C}, R(x,y)$. But $\exists x, \exists y \in \bar{C}, R(x,y)$ is an innocently excludable alternative in Q[34].

(125) **Fact (c).** If $\mathcal{Q}$ is UE, $\textsc{Exh}(\mathcal{Q} x, \exists y \in D, R(x,y))$ is equivalent to $(\mathcal{Q} x, \exists y \in D, R(x,y))$. (In other words, the *higher* prejacent is equivalent to the *lower* prejacent.)

We just need to exhibit maximally consistent set of alternatives which have no alternatives in common. For z in the domain D, let's construct the set of alternatives Alt_z which contains $\exists x, \exists y \in C, R(x,y)$ if z does not belong to C and $\mathcal{Q} x, \exists y \in C, R(x,y)$ if z does not belong to C. I claim that the negation of all alternatives in Alt_z is consistent with the prejacent for all z.

First off, note that any world where $\exists x, \exists y \in C, R(x,y)$ is false for all C which do not include z is a world where the relation R (restricted to $D_1 \times D$) is included in $D_1 \times \{z\}$. Reciprocally, note that whenever R is included in $D_1 \times \{z\}$, all the alternatives in Alt_z are automatically false. Indeed, in such a world, a statement like $\exists y \in C, R(x,y)$ is equivalent to $R(x,z)$ if z is in C and false otherwise. With this observation, the members of Alt_z simplify to $\exists x, \bot$ and $Q x, \bot$, which are all false ($\mathcal{Q}$ has existential entailments).

To prove consistency then, I just need to exhibit one world where the *lower* prejacent is true and R is included in $D_1 \times \{z\}$. This is easily done: take S a subset of size quant of D_1 and take a world where R is equal to $S \times \{z\}$ (and is thus included in $D_1 \times \{z\}$). In this world, $\mathcal{Q} x, \exists y \in D, R(x,y)$ is equivalent to $\mathcal{Q} x, x \in S$ and is true, since S's cardinality is in quant

To prove maximality, let us the previous observation that whenever the alternatives of Alt_z are false and z is in C, $\exists y \in C, R(x,y)$ is equivalent to $R(x,z)$. The alternatives which are not Alt_z are all equivalent to either $\exists x, R(x,z)$ and $\mathcal{Q} x, R(x,z)$, which are equivalent to $\exists x, x \in S$ and the *lower* prejacent respectively. Both of the latter are entailed by the prejacent so cannot be negated.

So all sets of alternatives Alt_z are maximal and consistent. But their intersection is empty: for any alternative $\mathcal{Q} x, \exists y \in C, R(x,y)$, one just need to find a z in D but not in C and this alternative will not be in Alt_z, and similarly for the other alternatives. No alternative is IE and the lower $\textsc{Exh}$ is vacuous.

(126) **Fact (d).** To the extent that the exhaustive participation inference is consistent with the higher prejacent, all *higher* alternatives are negatable and their negation, when conjoined to the prejacent, is equivalent to an exhaustive participation inference.

To show this, we simply need to show that all alternatives like $\textsc{Exh}\exists x, \exists y \in C, R(x,y)$ can be negated with the lower prejacent. By fact (b), this entails that all other alternatives are

[34] The negation of all alternatives is consistent and the prejacent and $\exists x, \exists y \in \bar{C}$ quantify over logically independent alternatives

negatable as well. Let us form this negation (127). I have grouped together alternatives which have complementary domains.

(127) $\neg\left(\exists x, \exists_C y, R(x,y) \wedge \neg\exists x, \exists_{\bar{C}} y, R(x,y)\right)$
$\wedge\neg\left(\exists x, \exists_{\bar{C}} y, R(x,y) \wedge \neg\exists x, \exists_C y, R(x,y)\right)$
$\wedge\neg\left(\exists x, \exists_{C_1} y, R(x,y) \wedge \neg\exists x, \exists_{\bar{C}_1} y, R(x,y)\right)$
$\ldots$

The crucial observation is that we can simplify these groupings, as follows[35]:

(128) $\neg\left(\exists x, \exists_C y, R(x,y) \wedge \neg\exists x, \exists_{\bar{C}} y, R(x,y)\right)$
$\wedge\neg\left(\exists x, \exists_{\bar{C}} y, R(x,y) \wedge \neg\exists x, \exists_C y, R(x,y)\right)$
$\leftrightsquigarrow$
$\left(\exists x, \exists_C y, R(x,y) \wedge \exists x, \exists_{\bar{C}} y, R(x,y)\right)$
$\wedge\left(\neg\exists x, \exists_{\bar{C}} y, R(x,y) \wedge \neg\exists x, \exists_C y, R(x,y)\right)$
$\leftrightsquigarrow$
$\left(\exists x, \exists_C y, R(x,y) \wedge \exists x, \exists_{\bar{C}} y, R(x,y)\right)$
$\wedge\left(\neg\exists x, \exists_D y, R(x,y)\right)$

Note that the second disjunct of (128) must be false if the prejacent is true. So together with the prejacent, the negation of the *higher* alternatives contributes the following inference:

(129) $\textsc{Exh}(\mathcal{Q}x, \exists y \in D, R(x,y))$
$\wedge\left(\exists x, \exists_C y, R(x,y) \wedge \exists x, \exists_{\bar{C}} y, R(x,y)\right)$
$\wedge\left(\exists x, \exists_{C_1} y, R(x,y) \wedge \exists x, \exists_{\bar{C}_1} y, R(x,y)\right)$
$\ldots$

This is exactly the conjunction of the prejacent with the exhaustive participation inferences. If this is consistent, then all *higher* alternatives are excludable

(130) $\textsc{Exh}(\mathcal{Q}x, \exists y \in D, R(x,y))$
$\wedge\forall y, \exists x, R(x,y)$
$\ldots$

Glossary

ι This meta-language symbol picks out from a meta-language predicate the maximal element (in terms of plural parthood) if it exists. ιsquirrels is thus the sum of all squirrels (since this sum is the maximal element in the extension of squirrels. This use goes against the common use of ι, as a uniqueness operator but it allows me to keep to Link's insight that there is no conceptual distinction between singular and plural definites. . 1, 2, 6, 7, 8, 9, 10, 11, 12, 14, 18, 22, 25, 36, 38, 40, 42, 43

$\prec$ This meta-language symbol translates "*is a plural part of*". This relation does not exclude singularities nor does it preclude non-proper parts. So all the following statements are

[35] The steps in this calculus: first, De Morgan's law; then, use or/and distributivity ; then simplify contradictory disjuncts

true: $X \prec X, x \prec x + y, x + y \prec x + y + z$. 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 18, 22, 25, 34, 36, 38, 39, 42, 43

cumulative reading The most salient reading of a sentence with at least two verbal arguments. The truth-conditions of these readings is typically paraphrased (although not always) by "*for all members of Y, there is a member of X such that ... and for all members of X, there is a member of Y, such that ...*". 1, 2, 3, 5, 6, 10, 39, 42, 43

dataset This refers to the set of the cumulative sentences in (9). These are the sentences which this chapter wants to give an account of. . 3, 4, 7, 8, 14, 19, 30, 32, 36, 43

doubly-distributive reading A reading of the form $\forall x, \forall y, R(x, y)$, where both arguments of a transitive verb are interpreted universally . 36, 39, 40, 41

ordinary cumulative sentence A sentence which contains nothing but plural-referring expressions. 2, 4, 41

plural-referring expression Any type e element which refers to a plural object, e.g. a definite plural, a conjunction of proper names, a conjunction of plural definites, etc. 1, 5, 6, 7, 8, 9, 10, 11, 12, 13, 16, 31, 36, 37, 38, 40, 41, 43, 44, 51

positive/negative environments In this chapter, an alias for upward and downward-entailing environments. In the future however, we will turn to conditionals, which despite being downward-entailing, license strong readings of plural expressions. When we get to this, it will be handy to have used a different term for the environments in which we find strong readings of plurals. . 5, 6, 8, 10, 11, 14, 16, 22, 36, 38, 39, 40, 43

prejacent When an operator like Exh applies to a constituent α, i.e. Exhα, I call α the prejacent of Exh. The term is used both for the object language (the linguistic constituent) and the meta-language (the formula to which meta-language Exh applies). . 23, 24, 25, 30, 33, 34, 47

truth-value gap Truth-value gap refers to any case where a sentence and its negation do not exhaust the space of alll possibilities ; some situations are not adequately described by either utterances. Presupposition, scalar implicatures and homogeneity are all types of truth-value gaps. . 5, 8, 11, 17

underlying meaning The underlying meaning of an utterance or lexical item is the meaning of this element prior to strengthening. In this chapter, this meaning is diagnosed by looking at the negative version of a sentence.. 5, 6, 16, 17, 18, 34, 38, 39

video-game example This refers to any of the variations on Schein (1993)'s example below:

(131) The ten video-games taught every quarterback $\left\{\begin{array}{l}\text{four new plays}\\ \text{less than four new plays}\\ \text{between four and six new plays}\end{array}\right.$

This example exhibits both a cumulative reading of *every* and a bare or modified numeral in its distributive scope. This example demonstrates that when it enters a cumulative reading, *every* still acts as a universal distributive quantifier for elements within its scope. . 3, 4, 8, 9, 10, 19, 35